

Honors Differential Equations

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Lecture 1: Introduction to Differential Equations

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Chapter 1

First-Order Differential Equations

1.1 Modeling via Differential Equations

In Differential Equations, we study equations relating functions & their derivatives. For example,

- If $m(t)$ denotes the amount of some radioactive material at some time t , then often we have

$$\frac{dm}{dt} = -km.$$

- If $y(t)$ is the height of a body above the ground, then a common model for the way the height changes over time is

$$\frac{d^2y}{dt^2} = -g.$$

Notation. Terminology that will be used throughout the course:

- one independent variable, usually t .
- one dependent variable which is a function of the independent variable, usually y , so we give $y(t)$.
- Parameters are constants that appear in the equation.

Definition 1.1: Order of a Differential equation

The order of a differential equation is the order of the highest derivative in that equation.

For example, in the case

$$\frac{d^2y}{dt^2} + y \frac{dy}{dt} + \left(\frac{dy}{dt}\right)^3 = 0,$$

the highest order derivative is order 2, so the equation is a second order differential equation.

Definition 1.2: System of Differential Equations

A system of differential equations involves two or more **dependent variables**.

For instance, if we have variables $y(t)$ and $z(t)$, then we might have

$$\frac{dy}{dt} = f(y, z, t), \quad \frac{dz}{dt} = g(y, z, t).$$

In this class, we only consider **only one independent variable**, since this would involve partial differential equations. However, we will see multiple dependent variables.

Note. Differential equations typically arise as models for some real world system. These models are often derived under some assumptions, which may not be satisfied in our particular circumstance, so solving the differential equation does not necessarily guarantee a solution in the real world.

For example,

$$\frac{d^2y}{dt^2} = -g$$

is a bad description of height change vs time for an object with significant air resistance.

Remark

Most differential equations cannot be solved explicitly; i.e. they do not have analytic solutions.

In response to this fact we will not only investigate analytic solutions to differential equations, but also solutions via qualitative methods and numerical methods.

Example

Let $P(t)$ be a population at time t . For example, some bacteria that periodically doubles in size. This is basically the opposite of the previously seen radioactive decay example, so we might think

$$\frac{dP}{dt} = rP,$$

where the parameter r is the birth rate. Since P is proportional to its derivative, we can describe it as the function

$$P(t) = Ce^{rt},$$

for $C \in \mathbb{R}$. Since we're modeling a population, at $t = 0$,

$$P(0) = C,$$

so in this case we take C to be the original population value at $t = 0$. So even though mathematically we can take $C \in \mathbb{R}$, the initial value requires us to have $C \in \mathbb{R}^+$.

Definition 1.3: Initial Value Problem

An initial value problem (IVP) is a differential equation with initial condition(s) describing the value of the dependent variable at some time t_0 .

Generally, we need the same number of initial conditions as the order of the differential equation if we want to solve for all the parameters.

Going back to 1.1, recall that as the population increases, resources may become scarce. We can modify our equation to account for this. Let

$$\frac{dP}{dt} = rP - kP^2 = kP \left(\frac{r}{k} - P \right).$$

For large populations, the second term will overwhelm the first term. We also know that if $\frac{r}{k} > P > 0$, then $\frac{dP}{dt} > 0$. However, if $P > \frac{r}{k}$, then $\frac{dP}{dt} < 0$. So for small populations, the population will grow, and for large populations, the population will begin to decrease.

This model is much harder to solve for P , so we can use qualitative analysis. We can draw a slope field drawn on the Pt -plane, wherein at each point (P, t) , we draw a line depicting the value of $\frac{dP}{dt}$ for that (P, t) . In this case, all slope values at

$$P = 0 \vee P = \frac{r}{k} =: P^*.$$

will be 0 (horizontal). As P approaches these values, $\frac{dP}{dt}$ tends to 0. For any point $P > P^*$, the slope will be negative. From this slope field, we know the behavior of the solution without finding the solution. So for $P_0 < P^*$,

$$\lim_{t \rightarrow \infty} P(t) = P^*.$$

And for $\tilde{P}_0 > P^*$,

$$\lim_{t \rightarrow \infty} \tilde{P}(t) = P^*.$$

— Remark —

The model

$$\frac{dP}{dt} = kP \left(\frac{r}{k} - P \right).$$

is known as the **logistic model**.

Lecture 2: Investigating the Logistic Model

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1.2 Separation of Variables

Solutions that are constant in time are known as “fixed points” or “equilibrium points” of the system. For example, for $P^* = P$, we have

$$\frac{dP}{dt} = kP(P^* - P) = 0.$$

And for $P_0 = 0$, we have a similar result.

This equation has an analytical solution.

$$\begin{aligned} \frac{\frac{dP}{dt}}{P(P^* - P)} &= k \\ \Rightarrow \int \frac{1}{P(t)(P^* - P(t))} \frac{dP}{dt} dt &= k \int dt = kt + C_1 \\ \text{Let } p = P(t) \Rightarrow dp &= \frac{dP}{dt} dt \\ \Rightarrow \int \frac{dp}{p(P^* - p)} &= kt + C_1 \\ \text{fuck partial fractions} \\ \Rightarrow \frac{1}{P^*} \ln \left| \frac{P(t)}{P^* - P(t)} \right| + C_2 &= kt + C_1 \end{aligned}$$

We have

$$\ln \left| \frac{P(t)}{P^* - P(t)} \right| = P^*kt + \tilde{C}.$$

$\tilde{C}$ is just a collection of constants. Take $t = t_0$ to solve for the initial condition problem.

$$\ln \left| \frac{P_0}{P^* - P_0} \right| = \tilde{C}.$$

$$\begin{cases} \text{Case 1: } 0 < P(t) < P^* \\ \text{Case 2:} \end{cases}$$

Investigation of case 1.

$$\ln \left| \frac{P(t)}{P^* - P(t)} \right| = \ln \left(\frac{P(t)}{P^* - P(t)} \right)$$

$$\implies \frac{P(t)}{P^* - P(t)} = e^{(rt + \tilde{C})} = e^{\tilde{C}} e^{rt}.$$

remember that $P^* = \frac{r}{k}$. We can finally solve for $P(t)$. Algebra gives

$$P(t) = \frac{P_0 P^* e^{rt}}{P^* + P_0 (e^{rt} - 1)}.$$

or something.

Say we harvest some fixed amount H of the population. We now have

$$\frac{dP}{dt} = kP(P^* - P) - H.$$

This is horrible to solve analytically. So we actually prefer to use a qualitative analysis on it, using a slope field.

Definition 1.4: Seperable Differential Equations

A separable differential equation is one of the form

$$\frac{dy}{dx} = f(y, t),$$

wherein we can separate the right hand side into two functions that only depend on one variable $f(y, t) = g(t)h(y)$.

Separable DE's are in principle always solvable. Taking definition 1.4, we can give

$$\frac{1}{h(y(t)) \frac{dy}{dx}} = g(t).$$

Integrating both sides, we have

$$\int \frac{1}{h(y(t)) \frac{dy}{dx}} dt = \int g(t) dt.$$

Let

$$\tilde{y} = y(t).$$

It follows that

$$d\tilde{y} = \frac{dy}{dx} dt.$$

We then substitute in and have

$$\int \frac{d\tilde{y}}{h(\tilde{y})} = \int g(t) dt.$$

Example

Suppose

$$\frac{dy}{dx} = t^2 y^2, \quad y(0) = 6.$$

Notice that y is a function of t . This is separable, so we have

$$\frac{1}{y^2} \frac{dy}{dt} = t^2.$$

We get

$$\int \frac{1}{y^2} dy = \int t^2 dt.$$

Solving, we have

$$-y^{-1} + C_1 = \frac{t^3}{3} + C_2.$$

Solving for y , we have

$$y = -\frac{1}{\frac{1}{3}t^3 + \tilde{C}}.$$

We can solve for $\tilde{C}$.

$$6 = \frac{1}{\tilde{C}}.$$

Trivial.

Example

Suppose

$$\frac{dy}{dt} = \frac{1}{y - t^2 y}, \quad y(0) = -4.$$

This is also separable since y factors:

$$y \frac{dy}{dt} = \frac{1}{1 - t^2}.$$

$$\begin{aligned} \int y \frac{dy}{dt} dt &= \int y dy \\ &= \frac{1}{2} y^2 + C_1 \\ \int \frac{1}{1 - t^2} dt &= \int \frac{\frac{1}{2}}{1 + t} + \frac{\frac{1}{2}}{1 - t} dt \\ &= \frac{1}{2} (\ln |1 + t| + \ln |1 - t|). \end{aligned}$$

Lecture 3: Continuing First Order ODEs

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Note - the previous example was actually

$$\frac{dy}{dt} = \frac{t}{y - t^2 y}.$$

This makes it way easier:

$$\begin{aligned} \int y dy &= \int \frac{t}{1 - t^2} dt. \\ \frac{1}{2} y^2 &= -\int \frac{1}{u} du. \\ \frac{y^2}{2} &= \ln |1 - t^2| + C. \end{aligned}$$

Solving from here is trivial.

An important problem is the **mixing problem**. Some container full of some solution, adding more stuff from the top while also draining out some stuff from the bottom. Rates of both can be diff. Stirring all the time to make sure things are mixed.

Example

Tank have 50gal of water. At $t = 0$, the tank contains pure water. Salty brine is poured into the tank at a rate of 3gal/min with a concentration of 1lb/gal of salt. Brine is drained from the tank at a rate of 3gal/min. How much salt is in the tank after 5 minutes?

Solution. Independent variable is time and dependent variable is amount of salt. Alternatively, salt concentration can be used (exercise to reader?)

Let $S(t)$ be the amount, in lbs, of salt at some time t . Since we are adding 1 lb per gallon and 3 gallons per minute, we add 3 lbs per minute. This is a constant positive rate of change of +3. We are then removing the amount of water being removed (3 gallons) times the concentration of salt in the tank, which is the amount of salt divided by 50 gallons.

$$\frac{dS}{dt} = 3 - 3 \left(\frac{S(t)}{50} \right).$$

The first term is adding salt, and the second term is removing salt. Simplifying, we get

$$\frac{dS}{dt} = 3 \left(1 - \frac{S(t)}{50} \right) = \frac{3(50 - S(t))}{50}.$$

This is separable.

$$\int \frac{1}{50 - S(t)} dS = \int \frac{3}{50} dt.$$

$$-\ln |50 - S(t)| = \frac{3t}{50} + C.$$

$$S(t) = 50 - e^{-C} e^{-\frac{3t}{50}}.$$

$$S(0) = 0 = 50 - e^{-C}.$$

Therefore, $e^{-C} = 50$, and $S(t) = 50 - 50 \exp \left(-\frac{3t}{50} \right)$. Plug in the time asked for.

$$S(5) = 50 \left(1 - e^{-\frac{15}{50}} \right) \text{ lbs.}$$

□

1.3 Euler's Method and Taylor's Theorem

Recall Taylor Polynomials. We know that an approximation for $f(t)$ about some $a \in \mathbb{R}$ is given by the Taylor Polynomial about a ,

$$f(t) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (t - a)^n.$$

A taylor polynomial of order n will be called $T_n(t)$.

— As Previously Seen —

Euler's Method:

Take an interval split into a grid of points t_j , separated by value Δt . Suppose $y(t_j)$ is known, at least approximately. Define $y(t_{j+1}) = y(t_j + \Delta t)$. This is approximately $y(t_j) + y'(t_j)\Delta t$. The derivative is usually given.

Given $\frac{dy}{dt} = f(y, t)$, $y(t_0) = y_0$,

- Pick a grid size Δt .
- Set $Y_0 = y_0$.
- Set $Y_1 = Y_0 + f(Y_0, t_0)\Delta t$.
- Set $Y_2 = Y_1 + f(Y_1, t_1)\Delta t$.
- Set $Y_{n+1} = Y_n + f(Y_n, t_n)\Delta t$.

In this case, $t_{n+1} = t_n + \Delta t$.

Theorem 1.1 (Taylor's theorem)

For some $f : \mathbb{R} \rightarrow \mathbb{R}$, then

$$f(t) = T_n(t) + R_{n+1}(t).$$

If $f(t)$ is $n + 1$ times continuously differentiable on $(a - \varepsilon, a + \varepsilon)$, then there exists a constant C_{n+1} such that $|f(t) - T_n(t)| = |R_{n+1}(t)| \leq C_{n+1} |t - a|^{n+1}$ for all $a - \varepsilon < t < a + \varepsilon$.

$$C_{n+1} := \max \left\{ \left| f^{(n+1)}[a - \varepsilon, a + \varepsilon] \right| \right\}.$$

Proof. By the FTC,

$$f(t) - f(a) = \int_a^t f'(s_1) \, ds_1 = \int_a^t f'(s_1) - f'(a) + f'(a) \, ds_1.$$

We thus have

$$f(t) = f(a) + \int_a^t f'(s_1) - f'(a) + f'(a) \, ds_1 = f(a) + \int_a^t f'(s_1) - f'(a) \, ds_1 + (t - a)f'(a).$$

Rearranging, we have

$$f(t) = f(a) + f'(a)(t - a) + \int_a^t f'(s_1) - f'(a) \, ds_1.$$

Define

$$R_2(t) := \int_a^t f'(s_1) - f'(a) \, ds_1.$$

From the FTC, we have

$$\int_a^t f'(s_1) - f'(a) \, ds_1 = \int_a^t \int_a^{s_1} f''(s_2) \, ds_2 \, ds_1.$$

We also know that for any real-valued function ϕ ,

$$\left| \int_{t_1}^{t_2} \phi(t) dt \right| \leq \int_{t_1}^{t_2} |\phi(t)| dt.$$

Thus,

$$R_2(t) \leq |R_2(t)| = \left| \int_a^t \int_a^{s_1} f''(s_2) ds_2 ds_1 \right| \leq \int_a^t \left| \int_a^{s_1} f''(s_2) ds_2 \right| ds_1.$$

Now let

$$M_2 := \max_{q \in [a, t]} |f''[a, t]|.$$

It follows naturally that

$$\int_a^t \left| \int_a^{s_1} f''(s_2) ds_2 \right| ds_1 \leq \int_a^t \int_a^{s_1} M_2 ds_2 ds_1 = M_2 \int_a^t \int_a^{s_1} ds_2 ds_1 = M_2 \int_a^t (s_1 - a) ds_1 = \frac{M_2}{2} (t - a)^2.$$

Therefore,

$$R_2(t) \leq M_2 \frac{(t - a)^2}{2!}$$

and thus,

$$f(t) \leq T_1(t) + M_2 \frac{(t - a)^2}{2!}$$

in an ε -neighborhood of a . This satisfies 1.1 for $\eta = 1$. Now we take an arbitrary $\eta = \xi - 1$. (this is equivalent to taking $\eta = \xi$ in the method of induction. I chose $\xi - 1$ for simplicity of notation.)

Suppose

$$f(t) = T_{\xi-1} + R_{\xi}(t) = \sum_{n=1}^{\xi-1} \left(\frac{f^{(n)}(a)}{n!} (t - a)^n \right) + \int_a^t \int_a^{s_1} \cdots \int_a^{s_{\xi-1}} f^{(\xi)}(s_{\xi}) ds_{\xi} \cdots ds_2 ds_1.$$

By the method of induction, it suffices to show

$$\int_a^t \int_a^{s_1} \cdots \int_a^{s_{\xi-1}} f^{(\xi)}(s_{\xi}) ds_{\xi} \cdots ds_2 ds_1 = \frac{f^{(\xi)}(a)}{\xi!} (t - a)^{\xi} + \int_a^t \int_a^{s_1} \cdots \int_a^{s_{\xi}} f^{(\xi+1)}(s_{\xi+1}) ds_{\xi+1} \cdots ds_2 ds_1,$$

and

$$\int_a^t \int_a^{s_1} \cdots \int_a^{s_{\xi}} f^{(\xi+1)}(s_{\xi+1}) ds_{\xi+1} \cdots ds_2 ds_1 \leq \max_{s_{\xi+1} \in [a, t]} |f^{(\xi+1)}[a, t]|.$$

For simplicity, we will denote $\tau + 1$ repeated integrals on the interval $[a, t]$ of some function $\phi(t)$ as

$$\left(J_{[a, t]}^{\tau+1} \phi \right) (t) := \int_a^t \int_a^{s_1} \cdots \int_a^{s_{\tau}} \phi(t) ds_{\tau} \cdots ds_1 dt.$$

We know that

$$\left(J_{[a, t]}^{\xi} f^{(\xi)} \right) (t) = \left(J_{[a, t]}^{\xi} g \right) (t)$$

for $g(t) = f^{(\xi)}(t) - f^{(\xi)}(a) + f^{(\xi)}(a)$. From this, by the FTC using steps seen previously, it follows that

$$R_{\xi}(t) = \left(J_{[a, t]}^{\xi} g \right) (t) = \left(J_{[a, t]}^{\xi+1} f^{(\xi+1)} \right) (t) + \left(J_{[a, t]}^{\xi} \right) (t) \cdot f^{(\xi)}(a).$$

We leave it as an exercise to the reader to show

$$\left(J_{[a,t]}^\xi\right)(t) = \frac{(t-a)^\xi}{\xi!}.$$

(hint: u-sub and induct on it!)

We then have

$$R_\xi(t) = \frac{f^{(\xi)}(t)}{\xi!}(t-a)^\xi + \left(J_{[a,t]}^{\xi+1} f^{(\xi+1)}\right)(t).$$

We can rewrite this as

$$\left(J_{[a,t]}^{\xi+1} f^{(\xi+1)}\right)(t) = \left(J_{[a,t]}^\xi \int_a^\xi f^{(\xi+1)}(s_{\xi+1}) ds_{\xi+1}\right)(t).$$

The first part of the proof has been clearly satisfied. Recall that for some real-valued $\phi : \mathbb{R} \rightarrow \mathbb{R}$,

$$\int_{t_1}^{t_2} \phi(t) dt \leq \left| \int_{t_1}^{t_2} \phi(t) dt \right|,$$

and

$$\left| \int_{t_1}^{t_2} \phi(t) dt \right| \leq \int_{t_1}^{t_2} |\phi(t)| dt.$$

Applying this first fact, followed by repeated applications of this second fact give us

$$\begin{aligned} \left(J_{[a,t]}^\xi \int_a^\xi f^{(\xi+1)}(s_{\xi+1}) ds_{\xi+1}\right)(t) &\leq \left| \left(J_{[a,t]}^\xi \int_a^\xi f^{(\xi+1)}(s_{\xi+1}) ds_{\xi+1}\right)(t) \right| \\ &\leq \left(J_{[a,t]}^\xi \int_a^\xi \left| f^{(\xi+1)}(s_{\xi+1}) \right| ds_{\xi+1}\right)(t) \end{aligned}$$

Define

$$M_{\xi+1} := \max_{s_\xi \in [a,t]} \left| f^{(\xi+1)}[a,t] \right|.$$

It follows naturally that

$$\left(J_{[a,t]}^\xi \int_a^\xi \left| f^{(\xi+1)}(s_{\xi+1}) \right| ds_{\xi+1}\right)(t) \leq \left(J_{[a,t]}^\xi \int_a^\xi M_{\xi+1} ds_{\xi+1}\right)(t) = M \left(J_{[a,t]}^{\xi+1}\right)(t) = \frac{M_{\xi+1}}{(\xi+1)!}(t-a)^{\xi+1}.$$

Therefore,

$$R_{\xi+1}(t) \leq \frac{M_{\xi+1}}{(\xi+1)!}(t-a)^{\xi+1}.$$

Thus, we are done. ■

Lecture 4: Alternative Derivation of Euler's Method

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$$\frac{dy}{dt} = f(y, t); \quad y(t_0) = y_0.$$

Goal: find an approximation to this initial value problem. Since continuous functions cannot be stored in a pc, we store approximations to the values of the function at a fixed set of points. On some interval $[t_0, t_k]$, we partition it into subintervals with endpoints $t_i = t_{i-1} + \Delta t$. Euler's method is to set $Y_0 = y_0$, then define $Y_{k+1} := Y_k + (\Delta t) \cdot f(Y_k, t_k)$.

Alternatively, we have

$$y_{t_k+1} = y(t_k + \Delta t) = y(t_k) + \Delta t(y'(t_k)) + R_2(\Delta t).$$

Basically, this is just a Taylor polynomial at t_k , which is near t_{k+1} . Rearranging, we have

$$y'(t_k)\Delta t = y(t_{k+1}) - y(t_k) - R_2(\Delta t).$$

Thus,

$$y'(t_k) = \frac{y(t_{k+1}) - y(t_k) - R_2(\Delta t)}{\Delta t}.$$

We remove the error to give the approximation

$$y'(t_k) \approx \frac{y(t_{k+1}) - y(t_k)}{\Delta t} \approx f(y(t_k), t_k).$$

The error (called the forward difference) is given as

$$\left| \frac{R_2(\Delta t)}{\Delta t} \right| \leq \frac{C_2 |\Delta t|^2}{|\Delta t|} \leq C_2 |\Delta t|.$$

We also don't use our actual value of $y(t_k)$, we use our value Y_{k+1} and Y_k .

1.4 Existence and Uniqueness of Solutions

Differential equations don't always have a solution, and may also have more than one solution. So how do we know if an initial value problem has a **unique solution**?

Theorem 1.2 (Picard)

Consider the initial value problem

$$\frac{dy}{dt} = f(y, t); \quad y(t_0) = y_0.$$

Suppose there exist constants $t_0 \in (a, b)$ and $y_0 \in (c, d)$ such that $\frac{\partial f}{\partial y}(y, t)$ and $f(y, t)$ are both continuous on $[a, b] \times [c, d]$. Then there exists some $\delta > 0$ such that the initial value problem has a **unique solution** $y(t)$ for all $t \in (t_0 - \delta, t_0 + \delta)$.

Proof.

$$\int_{t_0}^t \frac{dy}{ds}(s) ds = \int_{t_0}^t f(y(s), s) ds.$$

We have

$$\int_{t_0}^t \frac{dy}{ds}(s) ds = y(t) - y(t_0) = y(t) - y_0.$$

Thus,

$$y(t) = y_0 + \int_{t_0}^t f(y(s), s) ds.$$

We can inductively define a series of approximations. Our first approximation of y is that $y^{(0)}(t) = y_0$. Our next approximation is

$$y^{(1)}(t) = y_0 + \int_{t_0}^t f(y^{(0)}(s), s) ds.$$

Then we give

$$y^{(2)}(t) = y_0 + \int_{t_0}^t f(y^{(1)}(s), s) \, ds.$$

We continue to

$$y^{(n+1)}(t) = y_0 + \int_{t_0}^t f(y^{(n)}(s), s) \, ds.$$

By real anal, the approximations converge. ■

One important corollary of Picard's theorem, and in fact the most intuitive way to interpret Picard's theorem, is that any two solutions satisfying the theorem when plotted on the yt -plane, cannot intersect. Additionally, for autonomous equations, solutions cannot intersect on the fy -plane, since we can "shift" t_0 to make it an IVP at the point of intersection.

We also have a weaker theorem guaranteeing only the existence of a solution.

Theorem 1.3 (Existence Theorem)

Let $f(y, t)$ be continuous on $(y, t) \in N_1(t) \times N_2(y)$, for $N_n(x)$ a neighborhood $(x - \varepsilon_n, x + \varepsilon_n)$. For any $(y_0, t_0) \in N(t) \times N(y)$, there exists some function $y(t)$ and some ε such that $y(t)$ is defined for $(t_0 - \varepsilon, t_0 + \varepsilon)$ and solves the IVP

$$\frac{dy}{dt} = f(y, t), \quad y(t_0) = y_0.$$

Proof. not given. ■

Lecture 5: Fixing Missed Lecture Notes

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1.5 Equilibria and the Phase Line

I think at this point, it's useful to recap a working knowledge of the various graphs we can consider. First, we have the normal yt -plane. This is just a graph of $y(t)$. However, note that we can consider different initial values t_0 (generally $t_0 = 0$). Therefore, we may plot multiple $y(t)$ graphs on the same plane. All of these solutions solve the differential equation in the general sense, since an IVP induces a change only on the integration constants, and thus defines a specific equation within a class of general solutions.

The yt -plane may also be used as a slope field for non-autonomous equations. At any point (t, y) , we can define a value of the derivative $f(t, y)$, and draw a line pointing in a direction to indicate the value of the derivative. For example, a very large derivative would point almost completely upwards.

The fy -plane, where $f(t, y)$ is the derivative, is generally only used for autonomous equations, or for specific IVPs. It shows the value of the derivative vs the value of y .

Definition 1.5: Autonomous Differential Equation

An autonomous differential equation is an equation that depends only on y , and is constant with respect to t . We often write

$$\frac{dy}{dt} = f(y).$$

Drawing an entire slope field for an autonomous equation doesn't make much sense, since for any t , the value of $\frac{dy}{dt}$ will be the same, provided y is the same. Instead, we draw a single line, and place a node at each equilibrium point. For $y \in C^1$, the intermediate value theorem applies, and we can conclude for

$y \in (e_1, e_2)$ for e_1, e_2 equilibrium points, then either $y > 0$ or $y < 0$ is true for all y in the interval, provided no other equilibria exist in the interval. Thus, we can draw an arrow to indicate if the derivative is positive or negative on each section between equilibrium points.

This is a powerful tool for analyzing differential equations. By thm:1, a solution between equilibria will approach an equilibrium point, but never reach it. Thus, a phase line can tell us a lot about the behavior of solutions, provided we can find equilibria and some values of $\frac{dy}{dt}$.

This definition gives rise to a method of classifying equilibrium points.

Definition 1.6: Nodes, Sources, and Sinks

Let $\frac{dy}{dt} = f(y)$ be an autonomous differential equation with equilibrium point y^* , and no other equilibria within a neighborhood of y^* . Let $y \pm \varepsilon \in N(y^*)$.

- If $y^* + \varepsilon > 0$ and $y^* - \varepsilon < 0$, then we say y^* is a source.
- If $y^* + \varepsilon < 0$ and $y^* - \varepsilon > 0$, then we say y^* is a sink.
- If $y^* + \varepsilon > 0$ and $y^* - \varepsilon > 0$, then we say y^* is a node.
- If $y^* + \varepsilon < 0$ and $y^* - \varepsilon < 0$, then we say y^* is a node.

The graphical explanation of this naming scheme is that a source has the arrows pointing away from it on the phase line, a sink has arrows pointing towards it, and a node has arrows above and below it both pointing in the same direction.

Theorem 1.4 (Linearization)

Let $\frac{dy}{dt} = f(y)$ be an autonomous differential equation with equilibrium point y^* .

- If $f'(y^*) > 0$, then y^* is a source.
- If $f'(y^*) < 0$, then y^* is a sink.
- If $f'(y^*) = 0$, then more information is required to analyze the situation.

proofs r overrated yknow :)

1.6 Bifurcations

Oftentimes, our differential equations contain parameters. For example, the nuclear decay model

$$\frac{dP}{dt} = -\lambda P$$

has the parameter λ . However, there is not always a single fixed parameter we wish to study, so we also analyze what changes varying parameters induces on the solutions and behavior of differential equations.

Notation. We often use μ to denote a parameter, and we will often write $\frac{dy}{dt} = f_\mu(y)$ to denote an autonomous differential equation that depends on the parameter μ . To fix a value of $\mu = \mu'$, we write $f_{\mu'}(y)$.

Since each differential equation corresponds to a family of solutions, we call a function $f_\mu(y)$ a *one-parameter family*.

Consider the one-parameter family

$$f_\mu(y) = y^2 - 2y + \mu.$$

Notice that for $\mu = 0$, we have the equilibrium points

$$y(y - 2) = 0 \implies y \in \{0, 2\}.$$

However, for $\mu = 2$, we have the equilibrium points

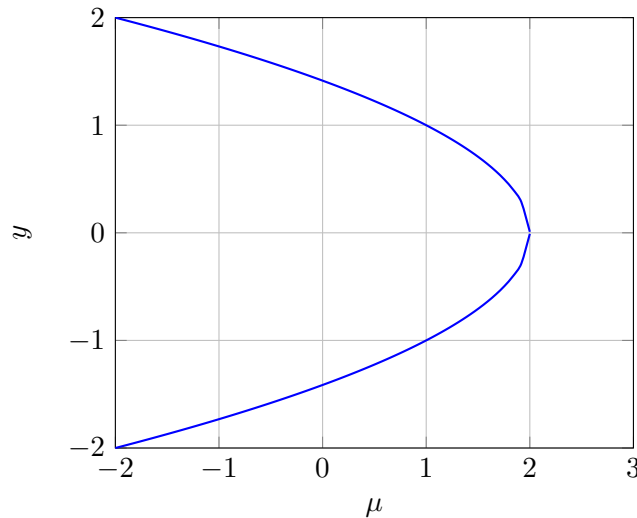
$$y^2 - 2y + 2 = 0 \implies y = \frac{2 \pm \sqrt{4 - 8}}{2} \in \mathbb{C} \setminus \mathbb{R}.$$

A small change in μ has deleted our equilibria for most purposes. We say a **bifurcation** has occurred.

Definition 1.7: Bifurcation

A bifurcation is a qualitative change in the nature of the equilibrium points (2 to 0 points). A point at which a bifurcation occurs is called a bifurcation value.

One way of qualitatively analyzing bifurcations is by considering the **bifurcation diagram**, aka a graph in the μy -plane of the equilibrium points. For example,



Notice how for $\mu > 2$, there are no equilibrium points, for $\mu = 2$, there's 1, and for $\mu < 2$, there's 2. Thus, $\mu = 2$ is a **bifurcation value**.

However, bifurcations do not usually happen.

Theorem 1.5 (Bifurcations)

Let $\frac{dy}{dt} = f_\mu(y)$ be a one-parameter family. Then a bifurcation occurs at $\mu = \mu_0$ if $f_{\mu_0}(y_0) = 0$ and $f'_{\mu_0}(y_0) = 0$.

1.7 Linear Equations

Definition 1.8: Linear Differential Equation

A differential equation is called linear if it is linear in y and its derivatives. That is, every linear differential equation is of the form

$$b(t) = \sum_{i=0}^n a_i(t)y^{(i)}(t).$$

We will generally study first order linear differential equations, which are of the form

$$\frac{dy}{dt} = a(t)y + b(t).$$

In general, linear differential equations are not separable. However, they are simple to analyze analytically. Our methods for doing so vary based on the specific equation, so it's useful to precisely classify each equation.

Definition 1.9: Homogeneous

Let

$$\frac{dy}{dt} = a(t)y + b(t).$$

If $b(t) \equiv 0$ for all t , then we say the differential equation is **homogeneous**. If $b(t)$ is nonzero, we say the equation is nonhomogeneous.

Linear differential equations act similarly to classic linear equations in many ways, and the next theorem formalizes some of these notions.

Theorem 1.6 (Linearity)

Let $\tilde{y}(t)$ solve the homogeneous equation

$$\frac{dy}{dt} = a(t)y.$$

Then any constant multiple $C \in \mathbb{R}$ of $\tilde{y}$ also solves the differential equation. That is,

$$\frac{dy}{dt} = a(t)\tilde{y} \implies \frac{dy}{dt} = a(t)C\tilde{y}.$$

Now consider the non-homogeneous equation

$$\frac{dy}{dt} = a(t)y + b(t).$$

- If y_h solves the associated homogeneous equation $y' = a(t)y$, and y_n is any solution to the nonhomogeneous equation, then $y_h + y_n$ solves the nonhomogeneous equation.
- If y_1, y_2 are two solutions to the nonhomogeneous equation, then $y_2 - y_1$ solves the associated homogeneous equation.

Solving homogeneous equations is fairly trivial, since they're separable by definition. Let

$$\frac{dy}{dt} = a(t)y.$$

It follows that

$$\begin{aligned} \implies \frac{1}{y} \frac{dy}{dt} &= a(t) \\ \implies \int \frac{1}{y} \frac{dy}{dt} dt &= \int a(t) dt \\ \implies \ln y &= A(t) + C \\ \implies y &= Ce^{A(t)}. \end{aligned}$$

Solving nonhomogeneous equations is more difficult, and a precise analytical method will be given in the next section.

1.8 Integrating Factors

Theorem 1.7 (Integrating Factors)

Let

$$\frac{dy}{dt} = a(t)y + b(t).$$

Then in principle, there always exists a solution $y(t)$ for the differential equation. Specifically,

$$y(t) = \left(e^{\int -a(t) dt} \right)^{-1} \int e^{\int -a(t) dt} b(t) dt.$$

From here, we can obviously account for an IVP.

Proof.

$$\frac{dy}{dt} = a(t)y + b(t).$$

Let $A(t)$ be any antiderivative of $-a(t)$. It follows that

$$e^{A(t)} \frac{dy}{dt} = a(t)e^{A(t)}y + b(t)e^{A(t)}.$$

Rearranging, we have

$$e^{A(t)} \frac{dy}{dt} - a(t)e^{A(t)}y = b(t)e^{A(t)}.$$

Notice that by the product rule,

$$\frac{d}{dt} \left(e^{A(t)}y(t) \right) = e^{A(t)}y'(t) + A'(t)e^{A(t)}y = e^{A(t)} \frac{dy}{dt} - a(t)e^{A(t)}y.$$

This is the same as the lefthand side. Thus,

$$\frac{d}{dt} \left(e^{A(t)}y(t) \right) = b(t)e^{A(t)}.$$

Integrate both sides, it follows that

$$e^{A(t)}y(t) = \int e^{A(t)}b(t) dt.$$

Rearranging, we have

$$y(t) = \left(e^{A(t)} \right) \int e^{A(t)} b(t) \, dt.$$

■

Chapter 2

First Order Systems

Lecture 6: Chapter 2 and Introducing Chapter 3

Wed 09 Oct 2024 12:25

2.1 Modeling via Systems

A system of differential equations is exactly what it sounds like. Consider some predator population $F(t)$ and prey population $R(t)$. We can describe their interaction as the system

$$\begin{aligned}\frac{dR}{dt} &= 2R - 1.2RF \\ \frac{dF}{dt} &= -F + 0.9RF.\end{aligned}$$

The $2R$ term depicts the exponential growth of prey in absense of predators, and the $-1.2RF$ depicts the negative predator-prey interaction. Conversely, the $-F$ term depicts the fact that predators die off if there is no prey to eat, and the $0.9RF$ term depicts the positive predator-prey interaction. The parameters are fairly arbitrary and depend on the species.

Chapter 3

Linear Systems

— As Previously Seen —

$$\frac{d\mathbf{Y}}{dt} = A\mathbf{Y}, \quad A(t) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \mathbf{Y} = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}.$$

For fixed point, Let $A\mathbf{v} = \lambda\mathbf{v}$ for some $\mathbf{v} \in \mathbb{R}^2$. Numbers λ that satisfy this are eigenvalues. Then

$$\mathbf{Y}(t) = e^{\lambda t}\mathbf{v}.$$

Worth remembering that

$$\lambda^2 + \text{tr}(A)\lambda + \det A = 0$$

is the characteristic polynomial in $M_2(\mathbb{R})$.

— Remark —

For $\lambda_1, \dots, \lambda_n$ eigenvalues of A , then $\text{tr}(A) = \sum_{i=1}^n \lambda_i$.

We have three cases to investigate,

1. $\lambda_+ \neq \lambda_-$, but $\lambda_+, \lambda_- \in \mathbb{R}$.
2. $\lambda_+ \neq \lambda_-$ and λ is complex. By Galois theory, $\lambda_+ = \lambda_-^*$.
3. $\lambda_+ = \lambda_-$. By Galois theory, both are real.

Since

$$\lambda_{\pm} = \frac{\text{tr}(A) \pm \sqrt{(\text{tr}(A))^2 - 4\det(A)}}{2},$$

then (2) arises only for

$$\det(A) > \frac{1}{4}(\text{tr}(A))^2.$$

Additionally, (1) arises for

$$\det(A) < \frac{1}{4}(\text{tr}(A))^2$$

and (3) arises for

$$\det(A) = \frac{1}{4}(\text{tr}(A))^2.$$

Now we have case (1a): $\lambda_- < 0 < \lambda_+$, (1b): $0 < \lambda_- < \lambda_+$, (1c): $\lambda_- < \lambda_+ < 0$. (1a) arises for

$$\text{tr}(A) > -\sqrt{\text{tr}^2(A) - 4\det A}.$$

It follows that $\det A < 0$. So for (1a), we have two straight line solutions $\mathbf{Y}^{(+)}(t) = e^{\lambda t} \mathbf{v}^{(+)}$, and likewise for $(-)$. The general solution $\mathbf{Y}(t)$ is then a linear combination of our basis:

$$\mathbf{Y}(t) = c^{(+)} \mathbf{Y}^{(+)}(t) + c^{(-)} \mathbf{Y}^{(-)}(t).$$

This gives rise to a *saddle* phase portrait. For an initial condition on $\mathbf{v}^{(-)}$, we collapse towards the origin. For $\mathbf{v}^{(+)}$, we diverge.

Now consider (1b). We have $\det A > 0$ and $\det A < \frac{1}{4} \text{tr}^2(A)$. All solutions go to infinity, and we call it a source. In (1c), they all go to 0, so we call it a sink.

Lecture 7: Continuation of Eigens

Fri 11 Oct 2024 12:53

Question: For complex eigenvalues, how do we find real-valued solutions?

Let $\mathbf{v}_+$ be an eigenvector. We have $\mathbf{v}_+ = \mathbf{u} + i\mathbf{w}$. We give $\lambda_+ = \alpha + i\beta$ and $\lambda_- = \lambda_+^*$. Thus, $\mathbf{v}_- = \mathbf{v}_+^*$. Then

$$\mathbf{Y}^{(+)}(t) = \exp(\lambda_+ t) \mathbf{v}_+ = e^{\alpha t} e^{i\beta t} (\mathbf{u} + i\mathbf{w}).$$

By euler's formula,

$$\begin{aligned} &= e^{\alpha t} (\cos(\beta t) + i \sin(\beta t)) (\mathbf{u} + i\mathbf{w}) \\ &= e^{\alpha t} (\cos(\beta t)\mathbf{u} + \cos(\beta t)i\mathbf{w} + i \sin(\beta t)\mathbf{u} + i \sin(\beta t)i\mathbf{w}) = e^{\alpha t} (\cos(\beta t)\mathbf{u} - \sin(\beta t)\mathbf{w} + i(\cos(\beta t)\mathbf{w} + \sin(\beta t)\mathbf{u})) \\ &= e^{\alpha t} (\cos(\beta t)\mathbf{u} - \sin(\beta t)\mathbf{w}) + ie^{\alpha t} (\cos(\beta t)\mathbf{w} + \sin(\beta t)\mathbf{u}). \end{aligned}$$

Question: For complex eigenvalues, how do we find real-valued solutions?

Let $\mathbf{v}_+$ be an eigenvector. We have $\mathbf{v}_+ = \mathbf{u} + i\mathbf{w}$. We give $\lambda_+ = \alpha + i\beta$ and $\lambda_- = \lambda_+^*$. Thus, $\mathbf{v}_- = \mathbf{v}_+^*$. Then

$$\mathbf{Y}^{(+)}(t) = \exp(\lambda_+ t) \mathbf{v}_+ = e^{\alpha t} e^{i\beta t} (\mathbf{u} + i\mathbf{w}).$$

By euler's formula,

$$\begin{aligned} &= e^{\alpha t} (\cos(\beta t) + i \sin(\beta t)) (\mathbf{u} + i\mathbf{w}) \\ &= e^{\alpha t} (\cos(\beta t)\mathbf{u} + \cos(\beta t)i\mathbf{w} + i \sin(\beta t)\mathbf{u} + i \sin(\beta t)i\mathbf{w}) = e^{\alpha t} (\cos(\beta t)\mathbf{u} - \sin(\beta t)\mathbf{w} + i(\cos(\beta t)\mathbf{w} + \sin(\beta t)\mathbf{u})) \\ &= e^{\alpha t} (\cos(\beta t)\mathbf{u} - \sin(\beta t)\mathbf{w}) + ie^{\alpha t} (\cos(\beta t)\mathbf{w} + \sin(\beta t)\mathbf{u}). \end{aligned}$$

We have $\mathbf{Y}^{(-)} = \mathbf{Y}^{(+)*}$ as well. Therefore, $\mathbf{Y}^{(+)} + \mathbf{Y}^{(-)} = \Re(\mathbf{Y}^{(+)}) = \Re(\mathbf{Y}^{(-)}) \in \mathbb{R}$. Since this is a linear combination of basis solutions, then we have the solution

$$\mathbf{Y}^{(1)} = e^{\alpha t} (\cos(\beta t)\mathbf{u} - \sin(\beta t)\mathbf{w}).$$

Additionally, if we take $\mathbf{Y}^{(+)} - \mathbf{Y}^{(-)} = 2i\Im(\mathbf{Y})$. Since $2i$ is a scalar, we have $\Im(\mathbf{Y})$ also a solution:

$$\mathbf{Y}^{(2)} = e^{\alpha t} (\cos(\beta t)\mathbf{w} + \sin(\beta t)\mathbf{u}).$$

These solutions are linearly independent, so we have a new basis over $\mathbb{R}$. This also implies our solution space is a subspace of the overall space. We have $\mathbf{Y}^{(-)} = \mathbf{Y}^{(+)*}$ as well. Therefore, $\mathbf{Y}^{(+)} + \mathbf{Y}^{(-)} = \Re(\mathbf{Y}^{(+)}) = \Re(\mathbf{Y}^{(-)}) \in \mathbb{R}$. Since this is a linear combination of basis solutions, then we have the solution

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$$\mathbf{Y}^{(2)} = e^{\alpha t} (\cos(\beta t)\mathbf{w} + \sin(\beta t)\mathbf{u}).$$

These solutions are linearly independent, so we have a new basis over $\mathbb{R}$. This also implies our solution space is a subspace of the overall space.

Lecture 8: Unsure

Wed 16 Oct 2024 12:20

We can give an example of this. Let

$$\frac{d\mathbf{Y}}{dt} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{Y}, \quad \mathbf{Y}(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

We have

$$\det \begin{bmatrix} -1 - \lambda & 1 \\ -1 & -\lambda \end{bmatrix} = \lambda^2 + \lambda + 1 = 0.$$

It follows that

$$\lambda = \frac{-1 \pm \sqrt{1-4}}{2} = -\frac{1}{2} \pm \frac{i\sqrt{3}}{2}.$$

We now have complex-valued eigenvalues.

$$\begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \lambda_+ \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}.$$

It will follow that

$$\mathbf{v} = \begin{bmatrix} -\lambda_+ \\ 1 \end{bmatrix}.$$

We then have

$$\mathbf{Y} = e^{\lambda_+ t} \mathbf{v} = e^{-\frac{t}{2}} e^{it\sqrt{3}/2} \mathbf{v}.$$

By Euler's formula,

$$\exp\left(-\frac{t}{2}\right) \exp\left(\frac{i\sqrt{3}}{2}t\right) = \exp\left(-\frac{t}{2}\right) \left(\cos\left(\frac{\sqrt{3}}{2}t\right) + i \sin\left(\frac{\sqrt{3}}{2}t\right) \right).$$

Multiplying through by $\mathbf{v}$, we have

$$= \exp\left(-\frac{t}{2}\right) \left(\begin{bmatrix} \frac{1}{2} \cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{\sqrt{3}}{2} \sin\left(\frac{\sqrt{3}}{2}t\right) \\ \cos\left(\frac{\sqrt{3}}{2}t\right) \end{bmatrix} + i \begin{bmatrix} \frac{1}{2} \sin\left(\frac{\sqrt{3}}{2}t\right) - \frac{\sqrt{3}}{2} \cos\left(\frac{\sqrt{3}}{2}t\right) \\ \sin\left(\frac{\sqrt{3}}{2}t\right) \end{bmatrix} \right) =: \mathbf{Y}^{(+)}.$$

We will have $\mathbf{Y}^{(-)} = (\mathbf{Y}^{(+)})^*$, and thus by linearity, $\mathbf{Y}^{(+)} + \mathbf{Y}^{(-)}$ solves the equation, and

$$\mathbf{Y}^{(+)} + \mathbf{Y}^{(-)} = \Re \mathbf{Y} = \begin{bmatrix} \frac{1}{2} \cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{\sqrt{3}}{2} \sin\left(\frac{\sqrt{3}}{2}t\right) \\ \cos\left(\frac{\sqrt{3}}{2}t\right) \end{bmatrix}.$$

Additionally, by linearity, $\mathbf{Y}^{(-)} - \mathbf{Y}^{(+)}$ gives us the solution

$$\Im \mathbf{Y} = \begin{bmatrix} \frac{1}{2} \sin\left(\frac{\sqrt{3}}{2}t\right) - \frac{\sqrt{3}}{2} \cos\left(\frac{\sqrt{3}}{2}t\right) \\ \sin\left(\frac{\sqrt{3}}{2}t\right) \end{bmatrix}.$$

This defines a basis for the solution space over $\mathbb{R}$, which is a subspace of the original solutions space over $\mathbb{C}$, since $\mathbb{R}$ is a subfield of $\mathbb{C}$.

This gives rise to a solution that spirals downwards towards the equilibrium solution $(0, 0)$.

Definition 3.1

If A has a pair of complex conjugate eigenvalues $\lambda = \lambda_r \pm i\lambda_I$, then

- If $\lambda_r < 0$, then we say we have a spiral sink.
- If $\lambda_r > 0$, then we say we have a spiral source.
- If $\lambda_r = 0$, then we say we have a center.

In all cases, the solution oscillates with an angular frequency λ_I .

Finally, consider the case of $\lambda_- = \lambda_+$. We could have either two eigenvectors or one eigenvector. For example,

$$\begin{bmatrix} 271 & \\ 0 & 2 \end{bmatrix}$$

has only $|0\rangle$ as an eigenvector

Lecture 9: Wrapping up

Fri 18 Oct 2024 12:27

Everywhere on the Trace-Determinant plane, there are two eigenvalues and two linearly independent eigenvectors if

$$\det A \neq \frac{1}{4} (\text{Tr } A)^2.$$

If we are on that parabola, however, there could be only one eigenvalue and eigenvector. For example,

$$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

has only $|0\rangle$ as an eigenvector, and only $\lambda = 2$ as an eigenvalue. We have at least the solution $\exp(\lambda t)|0\rangle$, but this is not a general solution.

Since $y = \exp(2t)y_0$, since the second equation is independent of x , we can plug this in and get

$$\frac{dx}{dt} = 2x + y_0 \exp(2t).$$

It follows that $x = x_0 \exp(2t) + y_0 t \exp(2t)$. Thus,

$$\mathbf{Y}(t) = \begin{bmatrix} x_0 e^{2t} + y_0 t e^{2t} \\ y_0 e^{2t} \end{bmatrix}.$$

Notice this is equivalent to

$$x_0 e^{2t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y_0 \begin{bmatrix} t e^{2t} \\ e^{2t} \end{bmatrix}.$$

This suggests that if A has only one eigenvalue λ and one eigenvector $\mathbf{v}$, then the “second solution” should be proportional to $t e^{\lambda t}$.

Suppose λ is the only eigenvalue of A and $\mathbf{v}$ is the corresponding eigenvector. We “guess” that there should be a solution of the form

$$\mathbf{Y}(t) = e^{\lambda t}(t\mathbf{v} + \mathbf{w}).$$

It remains to be shown what $\mathbf{w}$ is. We have

$$\frac{d\mathbf{Y}}{dt} = \lambda e^{\lambda t}(t\mathbf{v} + \mathbf{w}) + e^{\lambda t}\mathbf{v} = e^{\lambda t}A(t\mathbf{v} + \mathbf{w}).$$

We cancel the exponential, and have

$$\frac{d\mathbf{Y}}{dt} = \lambda(t\mathbf{v} + \mathbf{w}) + \mathbf{v} = A(t\mathbf{v} + \mathbf{w}).$$

Distributing, we have

$$\frac{d\mathbf{Y}}{dt} = \lambda t\mathbf{v} + \lambda\mathbf{w} + \mathbf{v} = At\mathbf{v} + A\mathbf{w}.$$

Since $\lambda\mathbf{v} = A\mathbf{v}$, we cancel $t\lambda\mathbf{v}$ with $tA\mathbf{v}$, and have

$$\lambda\mathbf{w} + \mathbf{v} = A\mathbf{w}.$$

Since $\mathbf{v}$ is known, this is a linear equation for $\mathbf{w}$. We claim this equation always has a solution.

Since

$$\frac{dy}{dt} = ay \implies y = y_0 e^{at},$$

why not give

$$\frac{d\mathbf{Y}}{dt} = A\mathbf{Y}$$

as

$$\mathbf{Y}(t) = e^{tA}\mathbf{Y}_0?$$

Recall that

$$e^{tA} = \sum_{n=0}^{\infty} \frac{t^n}{n!} A^n.$$

For example, consider the non-dampened harmonic oscillator

$$m \frac{d^2 x}{dt^2} = -kx.$$

If we let $v = \frac{dx}{dt}$, then

$$\frac{d}{dt} \begin{bmatrix} x \\ v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}.$$

We know $A^0 = I$, and $A^1 = A$. We also know

$$A^2 = \begin{bmatrix} -\frac{k}{m} & 0 \\ 0 & -\frac{k}{m} \end{bmatrix} = -\frac{k}{m} I.$$

Thus, we can compute $A^3 = -\frac{k}{m} A$. Additionally, $A^4 = \frac{k^2}{m^2} I$. In general,

$$A^n = \begin{cases} \left(-\frac{k}{m}\right)^{\frac{n}{2}} I, & n \in 2\mathbb{Z} \\ \left(-\frac{k}{m}\right)^{\frac{n-1}{2}} A, & n \notin 2\mathbb{Z} \end{cases}.$$

This can be proven easily with induction.

Thus, we have

$$\begin{aligned} e^{tA} &= \sum_{n=0}^{\infty} \frac{t^n}{n!} A^n = \sum_{\ell=0}^{\infty} \frac{t^{2\ell}}{(2\ell)!} A^{2\ell} + \sum_{\ell=0}^{\infty} \frac{t^{2\ell+1}}{(2\ell+1)!} A^{2\ell+1} = \left(\sum_{\ell=0}^{\infty} \frac{(-1)^\ell \left(t\sqrt{\frac{k}{m}}\right)^{2\ell}}{(2\ell)!} \right) I + \left(\sum_{\ell=0}^{\infty} \frac{(-1)^\ell \left(t\sqrt{\frac{k}{m}}\right)^{\ell+1}}{(2\ell+1)!} \right) A. \\ &= \cos\left(t\sqrt{\frac{k}{m}}\right) I + \sin\left(t\sqrt{\frac{k}{m}}\right) A. \end{aligned}$$

Therefore,

$$e^{At} = \begin{bmatrix} \cos\left(t\sqrt{\frac{k}{m}}\right) & \sin\left(t\sqrt{\frac{k}{m}}\right) \\ -\frac{k}{m}\sin\left(t\sqrt{\frac{k}{m}}\right) & \cos\left(t\sqrt{\frac{k}{m}}\right) \end{bmatrix}.$$

This does, in fact, solve the differential equation.

Note that

$$\sum_{n=0}^{\infty} \frac{t^n}{n!} A^n = e^{tA}$$

converges to a 2×2 matrix for any value of any matrix $A \in M_2$. Additionally,

$$\frac{d}{dt}(e^{tA}) = \frac{d}{dt} \sum_{n=0}^{\infty} \frac{t^n}{n!} A^n = \sum_{n=1}^{\infty} \frac{nt^{n-1}}{n!} A^n = \sum_{n=1}^{\infty} A \left(\frac{t^{n-1} A^{n-1}}{(n-1)!} \right) = A e^{tA} = e^{tA} A.$$

This implies

$$\frac{d}{dt}(e^{tA}) = A(e^{tA}),$$

and thus this solves the equation. NOTE THAT $e^A e^B \neq e^{A+B}$ in general.

Lecture 10: Inhomogeneous Continuation

Wed 23 Oct 2024 12:21

Linearity principles hold for inhomogeneous systems. In the case

$$\frac{d\mathbf{Y}}{dt} = A\mathbf{Y} + \mathbf{G}(t),$$

Let $\mathbf{Y} = (0, e^t)$. the general solution to the homogeneous case plus a particular solution to the inhomogeneous case will give the general solution for the inhomogeneous case. We can sometimes guess $\mathbf{Y} = Ae^{\lambda t}\mathbf{v}$ for some vector $\mathbf{v}$ and constants A, λ , but there is also a more analytic method.

Du Hamel's formula says the following. Let

$$\frac{d\mathbf{Y}}{dt} = A\mathbf{Y} + \mathbf{G}(t).$$

Rearrange it to give

$$\frac{d\mathbf{Y}}{dt} - A\mathbf{Y} = \mathbf{G}(t).$$

Choose $M(t)$ some matrix such that

$$M(t) \frac{d\mathbf{Y}}{dt} - M(t)A\mathbf{Y} = \frac{d}{dt}M(t)\mathbf{Y}.$$

It follows that

$$M(t) = e^{-At}.$$

Thus, let

$$e^{-At} \frac{d\mathbf{Y}}{dt} - e^{-At} A\mathbf{Y} = e^{-At} \mathbf{G}(t).$$

By the inverse of the product rule,

$$\frac{d}{dt}(e^{-At}\mathbf{Y}) = e^{-At}\mathbf{G}(t).$$

Integrating, we find

$$\mathbf{Y}(t) = e^{At} \int e^{-At} \mathbf{G}(t) dt,$$

since the inverse of the matrix e^{-At} is e^{At} . Another simple way of stating this is

$$\mathbf{Y}(t) = e^{tA}\mathbf{Y}(0) + e^{tA} \int_0^t e^{-tA} \mathbf{G}(t) dt.$$

Lecture 11: Inhomogeneous Systems

Mon 28 Oct 2024 12:26

Consider the dampened and driven harmonic oscillator

$$\frac{d^2x}{dt^2} + p \frac{dx}{dt} + qx = g(t).$$

Setting $\frac{dx}{dt} = v$, we know this is equivalent to a system.

Consider $g(t) = \gamma \cos(\mu t)$. By Euler's formula,

$$\Re \left(\frac{d^2x}{dt^2} + p \frac{dx}{dt} + qx \right) = \Re (\gamma e^{i\mu t}).$$

We want to then look for complex solutions to

$$\frac{d^2z}{dt^2} + p \frac{dz}{dt} + qz = \gamma e^{i\mu t}.$$

We can guess based on our previous analysis that $z_p(t) = be^{i\mu t}$ will solve the equation. Plugging in, we end up with

$$\gamma e^{i\mu t} = (-\mu^2 be^{i\mu t}) + (ip\mu be^{i\mu t}) + (be^{i\mu t}) \implies b = \frac{\gamma}{-\mu^2 + ip\mu + q},$$

provided the denominator is nonzero. Note that the denominator is 0 if $i\mu = \lambda$ for λ an eigenvalue of the system. Cases of this sort are known as resonance. Note that this only fails for $p = 0$, since we would require the real and imaginary components to both be 0, and $\mu, p, q \in \mathbb{R}$.

However, we still need to find the real component of the solution

$$\frac{\gamma}{-\mu^2 + ip\mu + q} e^{i\mu t}.$$

Recall the method of conjugation:

$$\frac{1}{a + bi} = \frac{a - ib}{(a + ib)(a - ib)} = \frac{a - ib}{a^2 + b^2}.$$

Applying this to our problem, we have

$$b = \frac{\gamma(-\mu^2 + q - ip\mu)}{(q - \mu^2)^2 + (p\mu)^2} = \frac{q\gamma - \gamma\mu^2}{(q - \mu^2)^2 + p^2\mu^2} - \frac{\gamma ip\mu}{(q - \mu^2)^2 + \mu^2 p^2}.$$

Therefore, by our previous arguments, we have the following set of basis solutions:

$$x_1(t) = \gamma \frac{q - \mu^2}{(q - \mu^2)^2 + (p\mu)^2} \cos(\mu t), \quad x_2(t) = \gamma \frac{p\mu}{(q - \mu^2)^2 + (p\mu)^2} \sin(\mu t).$$

Lecture 12: awa

Fri 01 Nov 2024 12:23

Lecture 13: Infinite Dimensional Systems

Mon 04 Nov 2024 12:57

We are going to look at partial differential equations herein. These equations often behave precisely like ODEs, but in infinite dimensions.

Suppose we have a wire that we are heating. We want to find the heat of the wire at some point along its length as a function of time. We then have a function u describing the heat, which is a function of both time t and position along the wire x . So $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ is the map $(t, x) \mapsto u$. We now have to use the partial derivatives of u . We claim

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2},$$

for κ some constant. Our initial condition will be some $u_0 = u(0, 0)$. Additionally, we also need to know boundary conditions. Suppose the wire has length ℓ and both ends of the wire are in an ice water bath. Our boundary conditions are then $u(0, t) = u(\ell, t) = 0$.

We can look for a straight line solution. That is, we want $u(x, t) = e^{\lambda t} v(x)$. For simplicity, let $\kappa = 1$. If we plug in, we have

$$\lambda e^{\lambda t} v(x) = e^{\lambda t} \frac{d^2 v}{dx^2}.$$

Canceling the exponentials, we have

$$\lambda v(x) = \frac{d^2 v}{dx^2}.$$

We now have reduced it to a problem we already know quite a lot about, but instead of initial conditions, we have **boundary conditions**. So $v(0) = v(\ell) = 0$.

Lecture 14: Insanity and More Infinite Dimensional Systems

Wed 06 Nov 2024 10:47

ASIDE: Let $\mathbf{F} = (x, y, z)$ be the solution to some 3 dimensional system of differential equations that orbits $\mathbb{R}^3$. Define a new $\xi\eta$ -plane that is orthogonal to $\mathbf{F}$ at some point. As t increases, the equation will loop back around and continue to intersect the $\xi\eta$ -plane, meaning we can define a discrete dynamical system $P(\xi, \eta)$ as simply the points $\mathbf{F}$ cuts the $\xi\eta$ -plane.

Change in notation: $v(x)$ is now called $\phi(x)$. Recall the question: are there solutions $\phi(x)$ such that $\phi(0) = \phi(\ell) = 0$, where ϕ solves

$$\lambda \phi(x) = \frac{d^2 \phi}{dx^2}?$$

We claim all nonzero solutions have $\lambda \in \mathbb{R}$. Suppose $\lambda, \phi(x)$ solve the problem. Note that if we take the complex conjugate, we find $\bar{\lambda}$ and $\bar{\phi}$ also solve the equation. If $\phi(0) = \phi(L) = 0$, then $\bar{\phi}(0) = \bar{\phi}(L) = 0$. Multiply through by $\bar{\phi}$:

$$\bar{\phi} \cdot \frac{d^2 \phi}{dx^2} = \lambda |\phi(x)|^2.$$

Integrating both sides, it follows that

$$\int_0^L \bar{\phi}(x) \cdot \frac{d^2 \phi}{dx^2} dx = \lambda \int_0^L |\phi(x)|^2 dx.$$

Using IBP on the left derivative, we have

$$\bar{\phi}(x) \frac{d\phi}{dx} \Big|_0^L - \int_0^L \frac{d\bar{\phi}}{dx} \frac{d\phi}{dx} dx = \bar{\phi}(x) \frac{d\phi}{dx} \Big|_0^L - \int_0^L \left| \frac{d\phi}{dx} \right|^2 dx.$$

Using our boundary terms, we know

$$\bar{\phi}(x) \frac{d\phi}{dx} \Big|_0^L = 0.$$

Our integrand is thus

$$\int_0^L \left| \frac{d\phi}{dx} \right|^2 dx.$$

Recalling our previous integral and solving for λ ,

$$\lambda = \frac{-\int_0^L \left| \frac{d\phi}{dx} \right|^2 dx}{\int_0^L |\phi(x)|^2 dx} \in \mathbb{R}.$$

We now claim that if $(\lambda_1, \phi_1), (\lambda_2, \phi_2)$ are solutions of our problem and $\lambda_1 \neq \lambda_2$, then $\langle \phi_1, \phi_2 \rangle = 0$, where we recall that for functions ϕ, ψ , the inner product is defined as

$$\langle \phi, \psi \rangle = \int_0^L \phi(x)\psi(x) dx.$$

Consider the inner product

$$\lambda_1 \langle \phi_1, \phi_2 \rangle = \int_0^L \lambda_1 \phi_1 \phi_2 dx.$$

Recall that we found

$$\frac{d^2 u}{dx^2} = \lambda \phi.$$

Thus,

$$\int_0^L \frac{d^2 \phi_1}{dx^2} \phi_2 dx.$$

Using IBP,

$$\left. \frac{d\phi}{dx} \phi_2(x) \right|_0^L - \int_0^L \frac{d\phi_1}{dx} \frac{d\phi_2}{dx} dx.$$

Recalling that ϕ_1, ϕ_2 solve the equation, the boundary terms go to 0. Using IBP again, we have

$$-\phi_1(x) \frac{d\phi_2}{dx} \Big|_0^L + \int_0^L \phi_1(x) \frac{d^2 \phi_2}{dx^2} dx.$$

The boundary terms once again go to 0, so plugging in $\lambda_2 \phi_2 = \frac{d^2 \phi_2}{dx^2}$, we have

$$\int_0^L \phi_1(x) \lambda_2 \phi_2(x) dx = \lambda \langle \phi_1, \phi_2 \rangle.$$

Therefore,

$$\lambda_1 \langle \phi_1(x), \phi_2(x) \rangle = \lambda_2 \langle \phi_1(x), \phi_2(x) \rangle.$$

Recall that we suppose $\lambda_1 \neq \lambda_2$. Therefore,

$$\langle \phi_1(x), \phi_2(x) \rangle = 0.$$

Now, we will solve for the eigenfunctions of this problem. Since λ is negative and real, write $\lambda = -\omega^2$. Then

$$\frac{d^2 \phi}{dx^2} = -\omega^2 \phi(x).$$

We know we will have two solutions, one in sin and one in cos. We could also solve this with an exponential using euler's formula, but that's boring. We have

$$\phi(x) = A \cos(\omega x) + V \sin(\omega x).$$

We still have to solve the boundary conditions, however. Letting $x = 0$, we have $\phi(0) = A$. Therefore, $A = 0$. Set $\phi(\ell) = 0$. We have

$$\phi(\ell) = B \sin(\omega \ell) = 0.$$

We must have $B \neq 0$, meaning $\omega\ell = n\pi$ for some $n \in \mathbb{N}$. Thus, we have as a solution $\phi(x) = B \sin(\omega x)$, provided

$$\omega = \frac{n\pi}{\ell}.$$

Therefore, we have infinite eigenvalues

$$\lambda_n = -\omega_n^2,$$

and infinite associated eigenfunctions

$$\phi_n(x) = B_n \sin\left(\frac{n\pi x}{\ell}\right).$$

Recall the original function

$$u(x, t) = e^{-\lambda t} \phi(x).$$

We have

$$u_n(x, t) = e^{-\left(\frac{n\pi}{\ell}\right)^2 t} \sin\left(\frac{n\pi x}{\ell}\right).$$

Although these solutions satisfy the boundary conditions and the heat equation, they don't necessarily satisfy the initial conditions **unless** the initial heat in the wire happens to be some sin function. However, even in 2D systems the solutions generally didn't solve the IVPs. So we took a linear combination of solutions and chose the appropriate constants. Similarly, we take a linear combination of all eigensolutions as our general solution:

$$u(x, t) = \sum_{n=1}^{\infty} B_n \exp\left(-\left(\frac{n\pi}{\ell}\right)^2 t\right) \sin\left(\frac{n\pi x}{\ell}\right).$$

In order to satisfy the initial conditions $u(x, 0)$, we would need

$$\sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{\ell}\right) \exp(0) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{\ell}\right) = u_0(x).$$

This is a Fourier series. Recall that conditions exist such that this is true. Multiply both sides by $\sin(m\pi x/\ell)$. We have

$$\sum_{n=1}^{\infty} \sin\left(\frac{m\pi x}{\ell}\right) B_n \sin\left(\frac{n\pi x}{\ell}\right) = u_0 \sin\left(\frac{m\pi x}{\ell}\right).$$

We will integrate both sides from 0 to ℓ .

$$\int_0^{\ell} \sum_{n=1}^{\infty} \sin\left(\frac{m\pi x}{\ell}\right) B_n \sin\left(\frac{n\pi x}{\ell}\right) dx = \int_0^{\ell} u_0 \sin\left(\frac{m\pi x}{\ell}\right) dx.$$

We will interchange the sum with the integral because that works for analytic series.

$$\sum_{n=1}^{\infty} \left(B_n \int_0^{\ell} \sin\left(\frac{n\pi x}{\ell}\right) \sin\left(\frac{m\pi x}{\ell}\right) dx \right) = \int_0^{\ell} u_0 \sin\left(\frac{m\pi x}{\ell}\right) dx.$$

The left integral is the inner product of our eigenfunctions, so it follows by orthogonality that the integral is 0 for all $n \neq m$, so we have

$$B_m \int_0^{\ell} \left(\sin\left(\frac{m\pi x}{\ell}\right) \right)^2 dx = \int_0^{\ell} u_0 \sin\left(\frac{m\pi x}{\ell}\right) dx.$$

Therefore,

$$B_m = \frac{\int_0^{\ell} u_0 \sin\left(\frac{m\pi x}{\ell}\right) dx}{\int_0^{\ell} \left(\sin\left(\frac{m\pi x}{\ell}\right) \right)^2 dx}.$$

We can simplify this formula somewhat. Using Euler's formula, $(\sin \theta)^2 = \frac{1}{2} - \frac{1}{2} \cos(2\theta)$. It follows that

$$\int_0^\ell \frac{1}{2} - \frac{1}{2} \cos\left(\frac{2n\pi x}{\ell}\right) dx.$$

Notice that each cosine is 0 after the integral, so we get $\ell/2$. Therefore,

$$B_m = \frac{2}{\ell} \int_0^\ell u_0 \sin\left(\frac{n\pi x}{\ell}\right) dx.$$

Let $u(x, t)$ be some function. Consider the partial differential equation

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2},$$

with initial condition $u(x, 0) = u_0$ and boundary conditions $u(0, t) = u(\ell, t) = 0$. The general solution to this system is

$$u(x, t) = \sum_{n=1}^{\infty} \exp\left(-\left(\frac{2n\pi}{\ell}\right)^2 t\right) \frac{4u_0}{\ell} \sin\left(\frac{2n\pi}{\ell} x\right).$$

$$u(x, t) = \sum_{n=1}^{\infty} \frac{4u_0 \sin\left(\frac{2n\pi}{\ell} x\right)}{\exp\left(\left(\frac{2n\pi}{\ell}\right)^2 t\right) \ell}.$$

Lecture 15: More Heat Equation

Fri 08 Nov 2024 12:26

However, how do we know some arbitrary $u_0(x)$ can be expanded in terms of $\sin x$ and $\cos x$ for all functions u_0 ? Note that

$$\int_0^\pi \sin^2(nx) dx = \frac{\pi}{2}.$$

We will regard this infinite sum as a limit of partial sums in increasing terms. Even with small numbers of terms, we get extremely good approximations to the original function. This even works with piecewise functions, you just end up breaking up the integral into multiple pieces. You need much more terms to approximate it well, but it does really well. In the limit to infinity, although the fourier series converges to the function, the maximum error itself remains nonzero in the limit somehow.

Now consider the same problem, but with new boundary conditions. Suppose the left end $x = 0$ is submerged in ice water, but the right is insulated such that no heat goes through it (flux 0). Fourier's law says the flux of heat is proportional to the rate of change in the temperature:

$$\Phi(x, t) \propto \frac{\partial u}{\partial x}, \quad \Phi(x, t) = -\lambda t \frac{\partial u}{\partial x}.$$

Thus we have

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = u_0(x), \quad u(0, t) = 0, \quad \Phi(L, t) = 0 \implies \frac{\partial u}{\partial x} = 0.$$

We have

$$\frac{d^2 \phi}{dx^2} = \lambda \phi(x), \quad \phi(0) = 0, \quad \phi'(L) = 0.$$

Note that, $\lambda < 0$ is still real, so define $\lambda = -\omega^2$.

$$\frac{d^2 \phi}{dx^2} = -\omega^2 \phi(x).$$

This has the general solution

$$\phi(x) = A \cos(\omega t) + B \sin(\omega t).$$

We now need to account for our boundary conditions. $\phi(0) = 0$ implies the cosine term drops out. Thus,

$$\phi(x) = B \sin(\omega t) \implies \phi'(x) = B\omega \cos(\omega t).$$

Plugging in our boundary condition $\phi'(L) = 0$, we have

$$B\omega \cos(\omega L) = 0.$$

Thus, either $\omega = 0$ or $\cos \omega L = 0$, implying $\omega L = \frac{(2n+1)\pi}{2}$. However, recall that $\omega = 0$ implies $\phi(x) = B \sin(\omega t) = 0$, so we cannot have $\omega = 0$. Therefore,

$$\omega L = \frac{(2n+1)\pi}{2} \implies \omega = \frac{(2n+1)\pi}{2L}.$$

We then have the solutions

$$u_n(x, t) = e^{-\left(\frac{(2n+1)\omega}{2L}\right)^2 t} B_n \sin\left(\frac{(2n+1)\pi}{2L} x\right).$$

Call

$$\lambda_n = -\left(\frac{(2n+1)\pi}{L}\right)^2, \quad \phi_n(x) = \sin\left(\frac{(2n+1)\pi}{2L} x\right).$$

We are choosing $B_n = 1$ here, but the conventional choice is $1/\ln(2L)$. Explanation left as an exercise. Also left as an exercise that

$$\langle \phi_n, \phi_m \rangle = 0, \quad n \neq m.$$

Now we have

$$u(x, t) = \sum_{n=1}^{\infty} B_n \exp\left(-\left(\frac{(2n+1)\pi}{2L}\right)^2 t\right) \sin\left(\frac{(2n+1)\pi}{2L} x\right).$$

Furthermore,

$$B_n = \frac{\int_0^L u_0(x) \phi_n(x) \, dx}{\int_0^L (\phi_n(x))^2 \, dx}.$$

Proof left as an exercise.

Chapter 4

Nonlinear Systems

Consider the Van der Pol equation:

$$\begin{aligned}\frac{dx}{dt} &= y \\ \frac{dy}{dt} &= -x + y - x^2.\end{aligned}$$

Lecture 16: Ven der Pol Equation

Mon 11 Nov 2024 12:23

Our first method of qualitative analysis is to focus on the origin. Take x, y small and notice that the x^2y term is very small, so we can drop it and only consider the linear system

$$\frac{d\mathbf{Y}}{dt} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

This is the **linearization** of this system. We will call this matrix J . Note that $\text{Tr } J = 1$ and $\det J = 1$. We would thus have a spiral source at the origin.

Consider the project-like competitive system

$$\frac{dx}{dt} = 2x - 4x^2 - 3xy, \quad \frac{dy}{dt} = y - 6y - xy.$$

We have fixed points at $x = 1/2$ and $y = 0$, and $x = 0$ and $y = 1/6$. Further, note $\frac{dx}{dt} = 0$ at

$$y = \frac{2}{3} - \frac{4}{3}x,$$

and $\frac{dy}{dt} = 0$ at

$$y = \frac{1}{6}x - \frac{1}{6}.$$

Setting these equal and solving for the equilibrium point, we find $(3/7, 2/21)$ is a fixed point. I don't know what the next step is but he says we need to use Taylor series.

Set $f(x, y) = 2x - 4x^2 - 3xy$. Recall that $f(3/7, 2/21) = 0$. By Taylor's theorem, we also know

$$f(x, y) = f\left(\frac{3}{7}, \frac{2}{21}\right) + \frac{\partial f}{\partial x}\left(\frac{3}{7}, \frac{2}{21}\right)\left(x - \frac{3}{7}\right) + \frac{\partial f}{\partial y}\left(\frac{3}{7}, \frac{2}{21}\right)\left(y - \frac{2}{21}\right) + \cdots.$$

If we are closed to the fixed point, then the further terms become small, so we ignore them. We know from our fixed point that

$$f(x, y) \approx 0 + \left(2 - \frac{24}{7} - \frac{6}{21}\right) \left(x - \frac{3}{7}\right) + \left(-\frac{9}{7}\right) \left(y - \frac{2}{21}\right) = -\frac{12}{7} \left(x - \frac{3}{7}\right) - \frac{9}{7} \left(y - \frac{2}{21}\right).$$

Now we have to set $g(x, y) = y - 6y^2 - xy$, $g_x(x, y) = -y$, $g_y = 1 - 12y - x$.

$$g(x, y) \approx \frac{\partial g}{\partial x} \left(\frac{3}{7}, \frac{2}{21}\right) \left(x - \frac{3}{7}\right) + \frac{\partial g}{\partial y} \left(\frac{3}{7}, \frac{2}{21}\right) \left(y - \frac{2}{21}\right).$$

$$g(x, y) \approx -\frac{2}{21} \left(x - \frac{3}{7}\right) - \frac{4}{7} \left(y - \frac{2}{21}\right).$$

Letting

$$\bar{x} = x - \frac{3}{7}, \quad \bar{y} = y - \frac{2}{21},$$

We have

$$\frac{d\bar{x}}{dt} = \frac{dx}{dt} = f(x, y) = -\frac{12}{7}\bar{x} - \frac{9}{7}\bar{y}.$$

$$\frac{d\bar{y}}{dt} = -\frac{2}{21}\bar{x} - \frac{4}{7}\bar{y}.$$

We have

$$\frac{d\bar{\mathbf{Y}}}{dt} = \begin{pmatrix} -\frac{12}{7} & -\frac{9}{7} \\ -\frac{2}{21} & -\frac{4}{7} \end{pmatrix} \bar{\mathbf{Y}}.$$

Call this matrix J again. We have $\text{Tr } J = -\frac{16}{7}$, $\det J = \frac{6}{7}$.

Now let's do Taylor series again, around $(0, 0)$. Let

$$f(x, y) = 2x - 4x^2 - 3xy.$$

We have

$$f(x, y) = f\left(\frac{1}{2}, 0\right) + \frac{\partial f}{\partial x} \left(\frac{1}{2}, 0\right) \left(x - \frac{1}{2}\right) + \frac{\partial f}{\partial y} \left(\frac{1}{2}, 0\right) y = -2 \left(x - \frac{1}{2}\right) - \frac{3}{2}y.$$

Letting $g(x, y) = y - 6y^2 - xy$, we have

$$g(x, y) = g\left(\frac{1}{2}, 0\right) + \frac{\partial g}{\partial x} \left(\frac{1}{2}, 0\right) \left(x - \frac{1}{2}\right) + \frac{\partial g}{\partial y} \left(\frac{1}{2}, 0\right) (y) = \frac{1}{2}y.$$

Once again, write

$$\bar{x} = x - \frac{1}{2}, \quad \bar{y} = y.$$

We have

$$\frac{d\bar{x}}{dt} = -2\bar{x} - \frac{3}{2}\bar{y}, \quad \frac{d\bar{y}}{dt} = \frac{1}{2}\bar{y}.$$

Write it again as

$$\frac{d\bar{\mathbf{Y}}}{dt} = J\bar{\mathbf{Y}}.$$

Since $\text{Tr } J, \det J < 0$, we have a saddle. But why are we calling the matrix J ? Well,

$$J = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix}$$

is the Jacobian. We can thus use the Jacobian to derive the linearization of a system about an equilibrium point, which can classify the fixed points.

Lecture 17: Function Spaces and Hilbert Spaces

Wed 13 Nov 2024 10:15

Recall the definition of a vector space.

Definition 4.1: Vector Space

A vector space is an abelian group V over a field $\mathbb{F}$, to which we associate the operation scalar multiplication that satisfies

- $\mathbf{u} + \mathbf{v} \in V$.
- $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
- There exists $\mathbf{0} \in V$ such that $\mathbf{0} + \mathbf{v} = \mathbf{v}$.
- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
- For all $\mathbf{v}$ there exists $-\mathbf{v}$ with $\mathbf{v} - \mathbf{v} = \mathbf{0}$.
- There exists $1 \in \mathbb{F}$ such that $1\mathbf{v} = \mathbf{v}$.
- $a(b\mathbf{v}) = (ab)\mathbf{v}$.
- $a(\mathbf{v} + \mathbf{u}) = a\mathbf{v} + a\mathbf{u}$.
- $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$.

To get our notions of direction and magnitude, we need to consider inner product spaces.

Definition 4.2: Inner Product

A map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ is called an inner product if it satisfies

- $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}$.
- $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$.
- $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ if and only if $\mathbf{u} = \mathbf{0}$.
- $\langle k\mathbf{u}, \mathbf{v} \rangle = k \langle \mathbf{u}, \mathbf{v} \rangle$.
- $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$.

Notice that in the complex case,

$$\langle \mathbf{u}, k\mathbf{v} \rangle = \bar{k} \langle \mathbf{u}, \mathbf{v} \rangle.$$

In the physics literature, they often change the fourth property to

$$\langle \mathbf{u}, k\mathbf{v} \rangle = k \langle \mathbf{u}, \mathbf{v} \rangle.$$

The canonical inner product in $\mathbb{C}^n$ is

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n u_i \overline{v_i}.$$

For real-valued continuous function spaces, we define

$$\langle f, g \rangle = \int_a^b f(x)g(x) \, dx.$$

For complex-valued function spaces, we define

$$\langle f, g \rangle = \int_a^b f(x) \overline{g(x)} dx.$$

Definition 4.3: Norm

Let V be an inner product space. Then define $\|\cdot\| : V \rightarrow \mathbb{F}$ as the map

$$\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}.$$

We call $\|\cdot\|$ a norm.

Definition 4.4: Distance

Let V be an inner product space with $\mathbf{u}, \mathbf{v} \in V$. The distance between any two vectors d is defined as

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

Definition 4.5: Orthogonal

Let V be an inner product space. Then we say $\mathbf{v}, \mathbf{u}$ are orthogonal if and only if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Exercise 4.1

Prove the pythagorean theorem in any inner product space. Let V be an inner product space with $\mathbf{u}, \mathbf{v} \in V$ and $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. Then

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

Proof.

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} + \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \overline{\langle \mathbf{u} + \mathbf{v}, \mathbf{u} \rangle} + \overline{\langle \mathbf{u} + \mathbf{v}, \mathbf{v} \rangle} \\ &= \overline{\langle \mathbf{u}, \mathbf{u} \rangle} + \overline{\langle \mathbf{v}, \mathbf{u} \rangle} + \overline{\langle \mathbf{u}, \mathbf{v} \rangle} + \overline{\langle \mathbf{v}, \mathbf{v} \rangle} \\ &= \overline{\langle \mathbf{u}, \mathbf{u} \rangle} + \overline{0} + \overline{0} + \overline{\langle \mathbf{v}, \mathbf{v} \rangle} \\ &= \overline{\langle \mathbf{u}, \mathbf{u} \rangle} + \overline{\langle \mathbf{v}, \mathbf{v} \rangle} \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2. \end{aligned}$$

■

Lecture 18: More Nonlinear Systems

Wed 13 Nov 2024 12:22

Recall our exploration of equilibria with taylor series. We will define some notation. Recall that

$$\frac{dx}{dt} = f(x, y), \quad \frac{dy}{dt} = g(x, y).$$

Let

$$\mathbf{F}(x, y) = \begin{pmatrix} f(x, y) \\ g(x, y) \end{pmatrix}.$$

Then

$$F(\mathbf{Y}) = F(\mathbf{Y}^*) + J(\mathbf{Y}^*)(\mathbf{Y} - \mathbf{Y}^*) + \dots.$$

If we set $\bar{\mathbf{Y}} = \mathbf{Y} - \mathbf{Y}^*$, then at an equilibrium point we have

$$\frac{d\bar{\mathbf{Y}}}{dt} = J(\mathbf{Y}^*)\bar{\mathbf{Y}} + \dots.$$

This is a linear constant coefficient equation. Usually this linearization of a vector field exhibits the same behavior near an equilibrium point as the actual vector field, but there are cases where it does not. For example, consider the system

$$\frac{dx}{dt} = y - x(x^2 + y^2), \quad \frac{dy}{dt} = -x - y(x^2 + y^2).$$

The only equilibrium point is $(0, 0)$, and the only linear terms are

$$\frac{dx}{dt} = y, \quad \frac{dy}{dt} = -x.$$

This produces a center at this equilibrium point. However, our original nonlinear system does not have a center. If this were the case, then the distance function from the origin would be constant, but it is not. Note that this system has $\text{Tr } A = 0$.

Fact: Let $\mathbf{Y}^*$ be an equilibrium point. If $\text{Tr}(J(\mathbf{Y}^*)) \neq 0$ and $\det(J(\mathbf{Y}^*)) \neq 0$, then the linearization of the system at the equilibrium point captures the behavior of nearby points of the full system.

Lecture 19: Nonlinear Qualitative Analysis

Fri 15 Nov 2024 12:22

We can consider bifurcations in nonlinear systems using the linearization. Consider the single parameter family

$$\frac{dx}{dt} = y - x^2, \quad \frac{dy}{dt} = x - a.$$

Note that if $\frac{dy}{dt} = 0$, then $x = a$. By the first equation, we then have $y = -a^2$ at an equilibrium point. Therefore, $(a, -a^2)$ is the only equilibrium point. We can compute

$$J(\mathbf{x}) = \begin{pmatrix} \frac{\partial \frac{dx}{dt}}{\partial x} & \frac{\partial \frac{dx}{dt}}{\partial y} \\ \frac{\partial \frac{dy}{dt}}{\partial x} & \frac{\partial \frac{dy}{dt}}{\partial y} \end{pmatrix} = \begin{pmatrix} -2x & -1 \\ 1 & 0 \end{pmatrix}.$$

Therefore, at the equilibrium point,

$$J(a, a^2) = \begin{pmatrix} -2a & -1 \\ 1 & 0 \end{pmatrix}.$$

Then $\text{Tr } J = -2a$, $\det J = 1$.

Plotting this on the trace-determinant plane, we end up with three equilibrium points: two where it crosses the parabola, and one where $a = 0$. Let

$$1 = \frac{1}{4}(-2a)^2.$$

Then $a = \pm 1$, so our bifurcations are at $a \in \{0, -1, 1\}$.

We want to make more global statements about the behavior of nonlinear systems. Given a system

$$\frac{dx}{dt} = f(x, y) \quad \frac{dy}{dt} = g(x, y),$$

we define the **nullclines** to be the curves in the xy -plane defined by the equations $f(x, y) = 0$ XOR $g(x, y) = 0$. The nullclines correspond to curves on which the vector field is either vertical or horizontal. Intersections between f and g nullclines give equilibrium points, HOWEVER f nullclines can intersect other f nullclines, and those do not generate equilibria.

Consider the systems

$$\frac{dx}{dt} = 2x - 4x^2 - 3xy, \quad \frac{dy}{dt} = y - 6y^2 - xy.$$

This has equilibria at $(0, 0)$, $(0, 1/6)$, $(1/2, 0)$, $(3/7, 2/21)$. These are a source, saddle, saddle, and sink, respectively. Also consider the system

$$\frac{dx}{dt} = 6x - x^2 - 4xy, \quad \frac{dy}{dt} = 5y - 2xy - 2y^2.$$

This has equilibria at $(0, 0)$, $(0, 5/2)$, $(6, 0)$, $(4/3, 7/6)$. These are a source, sink, sink, and saddle, respectively.

The first system has nullclines at the x - and y -axes, a line from $(0, 2/3)$ to $(1/2, 0)$, and a line from $(0, 1/6)$ to $(?, 0)$.

In a linear system with a saddle, we have two straight line solutions, one of which approaches the saddle and one of which diverges. In a **nonlinear** system with a saddle, we have two special solutions, one of which approaches the saddle as $t \rightarrow \infty$, and one as $t \rightarrow -\infty$. These solutions are known as **separatrices**.

Theorem 4.1 (Poincare-Bendixson)

Suppose $(x(t), y(t))$ solves the equation

$$\frac{dx}{dt} = f(x, y), \quad \frac{dy}{dt} = g(x, y),$$

and it is bounded for all $t \geq 0$. Then as $t \rightarrow \infty$, either (x, y) approaches an equilibrium point, or (x, y) approaches a periodic solution of the equation.

Definition 4.6: Periodic Solution

We say $(x(t), y(t))$ is a periodic solution of a system if there exists $T > 0$ such that $(x(T), y(T)) = (x(0), y(0))$.

Lecture 20: Poincare-Bendixson

Mon 18 Nov 2024 12:23

Consider the system

$$\begin{aligned} \frac{dx}{dt} &= x(2 - x - y). \\ \frac{dy}{dt} &= y(y - x^2). \end{aligned}$$

Then we have two x -nullclines, $x = 0$ and $y = 2 - x$. Notice that the vector field is always pointing up along $x = 0$. However, notice that in the first quadrant, the direction along the second nullcline depends on whether $y - x^2$ is positive or negative. Since $(1, 1)$ is on this line and the nullcline, we have upwards to the left of this point and downwards to the right.

Our y -nullclines are $y = 0$ and $y = x^2$. This second y -nullcline explains the behavior seen in the previous part. Notice that the vector field points right along the first until $x = 2$, then it points left. Furthermore, the vector field points right along the quadratic nullcline, until it hits $(1, 1)$, when it turns left.

From this analysis, we know the origin is a source, the equilibrium point at $(2, 0)$ is a sink, and the center point $(1, 1)$ is a saddle.

One thing we will note is that the linearization at $\mathbf{0}$ would not have revealed this information, but nullcline analysis was able to reveal it.

Somehow the hairy ball theorem is related to the index of equilibrium points.

Proof. (Proof of Theorem 4.1 Poincare-Bendixson). Recall the Jordan Curve theorem: let there be a simple closed curve C in a plane, where C is homeomorphic to $\mathbb{S}^1$. Then the loop divides the plane into an interior and an exterior region.

Given a solution $\mathbf{Y}$ of a system of differential equations

$$\frac{dx}{dt} = f(x, y), \quad \frac{dy}{dt} = g(x, y),$$

we say a point $\mathbf{x}$ is in the set $\omega^+(\mathbf{Y}(t))$ if there exists an infinite sequence $\{t_n\}_{n=1}^\infty$, with $t_n \rightarrow \infty$ as $n \rightarrow \infty$, and

$$\lim_{n \rightarrow \infty} \mathbf{Y}(t_n) = \mathbf{x}.$$

We can think of this as the set of limit points of the solution $\mathbf{Y}(t)$. We call this the omega limit set. An equivalent definition of the $\omega^+(\mathbf{Y}(t))$ set is the following.

$$\omega^+(\mathbf{Y}(t)) = \bigcap_{s \geq 0} \overline{\{\mathbf{Y}(t) \mid t > s\}}.$$

Note that the line is the closure of this set. Note that this contains the limit points but also the previous points, so taking s bigger removes those points.

Another fact is that if $\mathbf{Y}(t)$ remains in a bounded set for all $t \geq 0$, then $\omega^+(\mathbf{Y}(t))$ is nonempty.

The next claim we will prove.

Notation. Given a point $\mathbf{x}$, define $\phi^t(\mathbf{x})$ to be the solution of the system with initial conditions $\mathbf{x}$ at time t . This function is well-defined given that Picard's theorem holds.

We claim that if $\mathbf{x} \in \omega^+(\mathbf{Y}(t))$, then $\phi^t(\mathbf{x}) \in \omega^+(\mathbf{Y}(t))$.

Suppose $\phi^{t_1}(\phi^{t_2}(\mathbf{x}))$. Remark that since $\phi^t(\mathbf{x})$ outputs a POINT on the solution evaluated at t , this makes sense to talk about. By Picard's theorem, $\phi^{t_1}(\phi^{t_2}(\mathbf{x})) = \phi^{t_1+t_2}(\mathbf{x})$. Suppose $\mathbf{x} \in \omega^+(\mathbf{Y}(t))$. Then we know there exists $\{t_n\}_{n=1}^\infty$ such that

$$\lim_{n \rightarrow \infty} \phi^{t_n}(\mathbf{Y}(0)) = \mathbf{x}.$$

Now we have

$$\lim_{n \rightarrow \infty} \phi^{t_n+s}(\mathbf{Y}(0)) = \lim_{n \rightarrow \infty} \phi^s(\phi^{t_n}(\mathbf{Y}(0))) = \phi^s(\mathbf{x}).$$

This follows from a corollary of Picard's theorem, which is that the solution map is continuous. Now consider the following definition. We say a curve γ is transverse to a vector field $\mathbf{F}(\mathbf{r})$ if at every point $\mathbf{x}$ on γ , there is a tangent vector to γ that is linearly independent of $\mathbf{F}(\mathbf{x})$.

Fact 4: there cannot be an equilibrium point on a transverse curve. This means the vector cannot change directions on a transverse curve.

Suppose $\mathbf{Y} = (x(t), y(t))$ remains bounded but doesn't approach an equilibrium point. Fix $\mathbf{x} \in \omega^+(\mathbf{Y})$. Since this point is not an equilibrium point, pick a curve γ transverse to $\mathbf{Y} = (x(t), y(t))$

containing $\mathbf{x}$. Then there exists some sequence $\{t_n\}_{n=1}^\infty$ such that $\mathbf{Y}(t_n)$ approaches $\mathbf{x}$ as $n \rightarrow \infty$, and $\mathbf{Y}(t_n) \in \gamma$ for all n .

An orbit $\mathbf{Y}(t)$ must cross a transverse curve monotonically. That is, we can order the points along γ ; if we call this order $\prec$, then $\mathbf{Y}(t_n) \prec \gamma$ for some n implies the same for all n . We will prove this.

Fix a point $\mathbf{Y}(t_i)$, and also consider $\mathbf{Y}(t_{i+1})$. By an extension of the intermediate value theorem, if we are to increase between these points, then to decrease along γ we must cross our previous curve, which is impossible by Picard's theorem.

I think this is sufficient the rest is witchcraft. ■

Lecture 21: Hamiltonian Differential Equations

Wed 20 Nov 2024 13:03

Consider the mass on a spring, with position x governed by

$$m \frac{d^2 x}{dt^2} = -kx.$$

Note that in this case, we are ignoring friction. In physics, we talk a lot about energy. Specifically, the kinetic energy of a particle, or the potential energy of a particle. A conservative force, $\mathbf{F} = -\nabla f$ for some potential function f , conserves energy. For example, the value $V = \frac{1}{2}kx^2$ is the potential function for the restorative force of a spring $F = kx$. The total energy $H(x, v)$, where $v = \frac{dx}{dt}$ is velocity, is

$$H(x, v) = \frac{1}{2}mv^2 + \frac{1}{2}kx^2.$$

The first part is our general kinetic energy formula, and the second is our V . This quantity is called the **Hamiltonian**, and systems of this type are called Hamiltonian systems.

Lecture 22: Hamiltonian Systems

Mon 25 Nov 2024 12:22

Suppose $m = 1$ for simplicity. Then

$$\frac{d^2 x}{dt^2} = -kx$$

can be written as

$$\frac{dx}{dt} = p, \quad \frac{dp}{dt} = -kx.$$

Then

$$\frac{d\mathbf{Y}}{dt} = \begin{pmatrix} 0 & 1 \\ -k & 0 \end{pmatrix} \mathbf{Y}$$

for $\mathbf{Y} = (x, p)$. Recall again that the kinetic energy $K = \frac{1}{2}m \left(\frac{dx}{dt}\right)^2$. Substituting in $m = 1$ and $v = p$, we have

$$K = \frac{1}{2}p^2.$$

We also have the potential energy V . Recall again that for a force $\mathbf{F}$, the potential function f has $\mathbf{F} = -\nabla f$. Given that $\mathbf{F} = \frac{1}{2}kx^2\hat{\mathbf{i}}$, we have $V = \frac{1}{2}kx^2$. Then the total energy of the system, denoted as H for hamiltonian, is

$$H(x, p) = K + V = \frac{1}{2}p^2 + \frac{1}{2}kx^2.$$

We can now write for our specific system that

$$\frac{dx}{dt} = \frac{\partial H}{\partial p}, \quad \frac{dp}{dt} = -\frac{\partial H}{\partial x}.$$

Proposition 4.1

Suppose that $(x(t), p(t))$ solves the hamiltonian system

$$\frac{d\mathbf{Y}}{dt} = \begin{pmatrix} 0 & 1 \\ -k & 0 \end{pmatrix} \mathbf{Y}.$$

Then $H(x(t), p(t))$ is constant.

Proof. By the chain rule,

$$\frac{dH}{dt} = \frac{\partial H}{\partial x} \frac{dx}{dt} + \frac{\partial H}{\partial p} \frac{dp}{dt}.$$

Using our previous analysis, we have

$$\frac{dH}{dt} = \frac{\partial H}{\partial x} \frac{\partial H}{\partial p} - \frac{\partial H}{\partial p} \frac{\partial H}{\partial x} = 0.$$

■

We can now observe that for any system with a quantity $H(x, p)$ such that

$$\frac{dx}{dt} = \frac{\partial H}{\partial p}, \quad \frac{dp}{dt} = -\frac{\partial H}{\partial x},$$

that H will be constant along any solution.

Consider the more interesting example of the equations of motion of a pendulum. Let the mass m of the pendulum $m = 1$, and let it be attached by a massless bar to a frictionless pivot. Instead of using a Newtonian approach, we will use a Hamiltonian approach to find the equations of motion. Let θ be the independent variable, which will be the angle made with the vertical.

The kinetic energy is $K = \frac{1}{2}mv^2 = \frac{1}{2} \left(\frac{dx}{dt} \right)^2$. Since the velocity is tangential, this is

$$K = \frac{1}{2} \left(L \frac{d\theta}{dt} \right)^2.$$

Let $L = 1$ for simplicity.

The potential energy near the surface of the Earth of an object is $V = mgh$. We can write $h = L - L \cos \theta = 1 - \cos \theta$. Let $g = 1$. Then $V = 1 - \cos \theta$.

Let $p = \frac{d\theta}{dt}$. The hamiltonian $H(\theta, p) = K + V$ is

$$H(\theta, p) = \frac{1}{2}p^2 + (1 - \cos \theta).$$

We then have

$$\frac{d\theta}{dt} = \frac{\partial H}{\partial p} = p, \quad \frac{dp}{dt} = -\frac{\partial H}{\partial \theta} = \sin \theta.$$

This is a nonlinear system of differential equations. Notice it only has equilibrium points on the θ -axis, since we require $p = 0$ and $\sin \theta = 0$.

Remark that since $\theta \in [-\pi, \pi]$ and $p \in \mathbb{R}$, this is really a cylindrical system. However, we are considering it as a plane and just fixing θ in this interval. This means we better have a periodic function on the plane.

Now we want to classify the equilibrium points.

$$\mathbf{J} = \begin{pmatrix} 0 & 1 \\ -\cos \theta & 0 \end{pmatrix}.$$

At an equilibrium point, $-\cos \theta \in \{-1, 1\}$. Therefore, we will have a center at the origin. Except in general, we cannot conclude the behavior for a center. At $\theta = \pi$, the eigenvalues are ± 1 , which is a saddle.

However, for a Hamiltonian system, we can say a lot more. Consider a contour plot of H in the θp -plane, which is the plot of all curves where H is constant. Since H is constant for all solutions, then we know that H will remain on the same contour for every solution.

Consider the curve $h = H(p, \theta)$ for h small and positive? We have

$$\cos \theta = 1 - \frac{1}{2}\theta^2 + \frac{1}{4!}\theta^4 + \dots$$

We then have

$$h = \frac{1}{2}p^2 + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} x^n = \frac{1}{2}p^2 + \frac{1}{2}\theta^2 + \dots$$

The contours satisfying this equation are small distortions of circles, since $h = \frac{1}{2}p^2 + \frac{1}{2}\theta^2$ is a circle. We can now confirm that there is a center at the origin.

Proposition 4.2

If a given equilibrium point for a **hamiltonian** system is a center in its linearization, then it is a center.

This means that hamiltonian solutions result in an effective decrease in dimension. Solutions can only wander along individual contours of a hamiltonian.

Now consider the hamiltonian at π . We have $H(0, \pi) = 1 - \cos \pi = 2$. What does the contour look like here?

We can rearrange to find

$$p^2 = 2(1 + \cos \theta).$$

We then have

$$p = \pm \sqrt{2 + 2 \cos \theta}.$$

If θ is small, then $\cos \theta \approx 1 - \frac{1}{2}\theta^2$. We then have

$$p = \sqrt{2 + 2 - \theta^2} = \sqrt{4 - \theta^2}.$$

The curve then has a maximum of 2, and it curves downwards to the two saddlepoints. The saddle's qualitative behavior comes up when $H > 2$.

One final remark for Hamiltonian systems is the following. If $\mathbf{x}$ is an equilibrium point of a hamiltonian system

$$\frac{dx}{dt} = \frac{\partial H}{\partial p}, \quad \frac{dp}{dt} = -\frac{\partial H}{\partial x},$$

then the Jacobian is

$$\mathbf{J} = \begin{pmatrix} \frac{\partial^2 H}{\partial x \partial p}(\mathbf{x}) & \frac{\partial^2 H}{\partial p^2}(\mathbf{x}) \\ -\frac{\partial^2 H}{\partial x^2}(\mathbf{x}) & -\frac{\partial^2 H}{\partial x \partial p}(\mathbf{x}) \end{pmatrix}.$$

We then have $\text{Tr } \mathbf{J} = 0$, so we will have either a saddle or a center.

Lecture 23: Power Series

Mon 02 Dec 2024 12:22

Suppose the power series

$$\sum_{n=0}^{\infty} a_n t^n, \quad \sum_{n=0}^{\infty} b_n t^n$$

both have positive radii of convergence, and suppose

$$\sum_{n=0}^{\infty} a_n t^n = \sum_{n=0}^{\infty} b_n t^n$$

for all $|t| < r$ for some $r > 0$. Then $a_n = b_n$ for all $n \in \mathbb{N}$. This follows rather simply from the fact that they are equal at $t = 0$. If $t = 0$, then the power series become $a_0 = b_0$. We can then take the derivative with respect to t , and plug in 0 to obtain $a_1 = b_1$. By inducting on this argument, we can show $\{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=0}^{\infty}$. By definition, the corresponding power series are equal.

A corollary of this is the following. Suppose

$$\sum_{n=0}^{\infty} b_n t^n = 0$$

for all $|t| < r$. Then we have $a_n = 0$ for all n . Suppose further that we have the derivative of some power series

$$\frac{d}{dt} \left(\sum_{n=0}^{\infty} a_n t^n \right) = \sum_{n=0}^{\infty} n a_n t^{n-1}.$$

Since it's often better to express both series in terms of t^n rather than t^{n-1} , we let $m = n - 1$ and represent this as

$$\sum_{n=1}^{\infty} n a_n t^{n-1} = \sum_{m=0}^{\infty} (m+1) a_{m+1} t^m.$$

Since m is just a dummy variable, we can plug in n for m and the series remains consistent. We then have

$$\frac{d}{dt} \left(\sum_{n=0}^{\infty} a_n t^n \right) = \sum_{n=0}^{\infty} (n+1) a_{n+1} t^n.$$

We can then take the second derivative and obtain

$$\sum_{n=0}^{\infty} n(n+1) a_{n+2} t^n.$$

In general, we will then have

$$\frac{d^k}{dt^k} \left(\sum_{n=0}^{\infty} a_n t^n \right) = \sum_{n=0}^{\infty} \left(a_{n+k} t^n \prod_{i=1}^k (n+i) \right).$$

Now consider the next fact. Consider the IVP

$$\frac{dy}{dt} = g(t)y + h(t), \quad y(0) = y_0.$$

If $g(t), h(t)$ have power series with radii of convergence $R > 0$, then the IVP has a solution $y(t)$ in the form of a power series with radius of convergence $R > 0$. One interesting thing to remark here is that by the previous fact, $a_0 = y_0$.

Another fact is the following. Suppose $r(t), p(t), q(t), h(t)$ all have power series' with positive radii of convergence. If $r(t) \neq 0$, then the differential equation

$$r(t) \frac{d^2 y}{dt^2} + p(t) \frac{dy}{dt} + q(t)y = h(t)$$

has a power series solution with positive radius of convergence. We will often write this as the equation

$$\frac{d^2y}{dt^2} + \tilde{p}(t) \frac{dy}{dt} + \tilde{q}(t)y = \tilde{h}(t),$$

where

$$\tilde{p}(t) = \frac{p(t)}{r(t)}.$$

There are, however, examples where $r(0) \neq 0$. This makes our representation fail.

Lecture 24: More Power Series

Wed 04 Dec 2024 10:11

Consider Legendre's Equation, which only has power series solutions:

$$(1 - t^2) \frac{d^2y}{dt^2} - 2t \frac{dy}{dt} + \lambda y = 0.$$

We want power series solutions

$$y(t) = \sum_{n=0}^{\infty} a_n t^n.$$

We primarily want solutions that are bounded for all $t \in [-1, 1]$ for reasons due to the nature of the equation itself. Our usual method is to start by differentiating.

$$\frac{dy}{dt} = \sum_{n=0}^{\infty} n a_n t^{n-1}.$$

Ordinarily we shift the index, but we will not do this because the term in the equation has a $-2t$ on it:

$$-2t \frac{dy}{dt} = \sum_{n=0}^{\infty} -2n a_n t^n.$$

If we differentiate again, we have

$$\frac{d^2y}{dt^2} = \sum_{n=0}^{\infty} n(n-1) a_n t^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} t^n.$$

We will actually use both of these forms by distributing out:

$$(1 - t^2) \frac{d^2y}{dt^2} = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} t^n - \sum_{n=0}^{\infty} n(n-1) a_n t^n.$$

We then have

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} t^n - \sum_{n=0}^{\infty} n(n-1) a_n t^n - \sum_{n=0}^{\infty} 2n a_n t^n + \sum_{n=0}^{\infty} \lambda a_n t^n = 0.$$

We can combine all these sums due to the indices being the same.

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} t^n + (-n(n-1) - 2n + \lambda) a_n t^n = 0.$$

Therefore, by combining like terms we know each term is 0 for all t :

$$(n+2)(n+1) a_{n+2} - (n(n+1) - \lambda) a_n = 0.$$

We then get the recursion relation

$$a_{n+2} = \frac{n(n+1) - \lambda}{(n+2)(n+1)} a_n.$$

We thus need two base cases, a_0 and a_1 . For a_2 , we have

$$a_2 = \frac{-\lambda}{2} a_0, \quad a_4 = \frac{6 - \lambda}{4 \cdot 3} a_2 = \frac{(2 - \lambda)(-\lambda)}{4!} a_0.$$

Inducting, we have

$$a_{n+2} = \frac{(n(n+1) - \lambda)((n-2)(n-1) - \lambda) \cdots (6 - \lambda)(-\lambda)}{(n+2)!} a_0.$$

We can do a similar analysis to identify a_n for odd n . Unfortunately, all these series diverge if they have infinitely many solutions, meaning we require all the terms after some specific term to vanish. We can remove all terms beyond the n th term by setting $\lambda = n(n+1)$. This gives a function with finitely many terms, aka a polynomial. So for any $\lambda_n = n(n+1)$, there exists a polynomial solution, known as the Legendre polynomial P_n of order n . If we work with the odd eigenvalues, we choose $a_0 = 0$ to get rid of the even terms for some reason, and vice versa.

Lecture 25: Sturm-Liouville Problems

Fri 06 Dec 2024 12:20

Consider the problem

$$-\frac{d}{dt} \left[p(t) \frac{dy}{dt} \right] + q(t)y(t) = \lambda r(t)y(t),$$

where p, q, r are given functions and λ is constant. let us have symmetric boundary conditions

$$A_1 y(a) + A_2 y'(a) = 0, \quad B_1 y(b) + B_2 y'(b) = 0$$

for $(A_1, A_2)(B_1, B_2) \neq \mathbf{0}$ and $a < b$. These equations arise very naturally in the theory of PDEs.

If $p, q, r \in C^1$ and if $p(t)$ and $r(t)$ are strictly greater than 0 on $[a, b]$, then we say that the equation plus these boundary conditions are a regular Sturm-Liouville boundary value problem. Every regular problem has every solution having a corresponding value $\lambda \in \mathbb{R}$, and there exists an infinite sequence $\lambda_1 < \lambda_2 < \dots$ with

$$\lim_{n \rightarrow \infty} \lambda_n = +\infty$$

such that the problem has a solution $(\lambda_n, y_n(t))$ for each $n \in \mathbb{Z}^+$. There exist no solutions to $\lambda \in \mathbb{C} \setminus \mathbb{R}$. Finally, every solution $y(t)$ is weakly-orthogonal with the integral

$$\int_a^b r(t) y_n(t) y_m(t) dt = 0 \forall m \neq n.$$

Legendre's Equation is of the form of a Sturm-Liouville problem, but since it has no boundary conditions, it's not a **regular** Sturm-Liouville BVD.

$$-\int_a^b y(t) \frac{d}{dt} \left(p(t) \frac{dy}{dt} \right) dt + \int_a^b q(t) (y(t))^2 dt = \lambda \int_a^b r(t) (y(t))^2 dt.$$

Using integration by parts on the first integral to combine the first and second integrals, and the boundary terms vanish (EFY).

$$\int_a^b p(t) \left(\frac{dy}{dt} \right)^2 q(t) (y(t))^2 dt = \lambda \int_a^b r(t) (y(t))^2 dt.$$

We can solve for λ and write out a messy af equation.

Define

$$\Lambda(x, y) = \frac{ax^2 + 2bxy + dy^2}{x^2 + y^2}.$$

In this case, $a, b, d \in \mathbb{R}$ are arbitrary. We want to maximize the function, so we set $\nabla \Lambda = \mathbf{0}$. We have

$$\frac{\partial \Lambda}{\partial x} = \frac{2ax + 2by}{x^2 + y^2} - \frac{2x(ax^2 + 2bxy + dy^2)}{(x^2 + y^2)^2} = \frac{2ax + 2by}{x^2 + y^2} - \frac{2x}{x^2 + y^2} \Lambda(x, y).$$

Set this equal to 0 and multiply through by $x^2 + y^2$. We have

$$2ax + 2by - 2x\Lambda(x, y) = 0.$$

We get

$$x\Lambda(x, y) = ax + by.$$

We also end up with

$$\frac{\partial \Lambda}{\partial y} = 0 \implies y\Lambda(x, y) = bx + dy.$$

This is very similar to the eigenvalue equation for a two-dimensional matrix:

$$\begin{pmatrix} ax + by \\ bx + dy \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}.$$

This generalizes. One more thing: note that

$$\Lambda(x, y) = \frac{\langle v, Av \rangle}{\|v\|^2}.$$

We can try doing this to our Sturm-Liouville problem. Now recall the previous representations of the eigenvalues of our Sturm-Liouville problems. We have a similar representation:

$$\lambda = \frac{\int_a^b \left(p(t) \left(\frac{dy}{dt} \right)^2 + q(t)(y(t))^2 \right) dt}{\int_a^b r(t)(y(t))^2 dt}.$$

This is basically the same, we have our inner products, except our variables are now functions. To find the eigenvalues, we need to minimize this function of functions. This leads to the calculus of variations.

Consider two points $(a, A), (b, B)$. We want to find the shortest line between them. Let C be a smooth curve given by the function $y(t)$. We can find the arclength using the formula

$$L(y) = \int_a^b \sqrt{1 + \left(\frac{dy}{dt} \right)^2} dt.$$

Suppose $\bar{y}(t)$ is the curve that minimizes $L(y)$. Let $\phi(t)$ be a continuously differentiable function with $\phi(a) = \phi(b) = 0$.

Note that $y_\varepsilon(t) = \bar{y}(t) + \varepsilon\phi(t)$ satisfies

$$y_\varepsilon(a) = \bar{y}(a) + \varepsilon\phi(a) = A.$$

By the same idea, $y_\varepsilon(b) = B$. So for any ε , this function y_ε satisfies the boundary conditions. Since $\bar{y}$ is the minimum, $L(y_\varepsilon) > L(\bar{y})$ for all ε . Define $\mathcal{L}(\varepsilon) = L(y_\varepsilon)$. We now have a function of one real variable with a minimum at $\varepsilon = 0$.

We can take the derivative. We have the derivative at 0

$$\frac{d\ell}{d\varepsilon}(0) = 0.$$

Ok now this appears from our previous formula.

$$\ell(\varepsilon) = \int_a^b \sqrt{1 + \left(\frac{d\bar{y}}{dt} + \varepsilon \frac{d\phi}{dt} \right)^2} dt.$$

Lecture 26: Calculus of Variations

Mon 09 Dec 2024 12:21

Recall: if $\phi(t)$ satisfies $\phi(a) = \phi(b) = 0$, then $y_\varepsilon(t) = \bar{y}(t) + \varepsilon\phi(t)$ satisfies the boundary conditions for all ε . As a consequence, since $\bar{y}$ minimizes the length, we see that

$$L(y_\varepsilon(t)) \geq L(\bar{y}(t)).$$

Define

$$\ell(\varepsilon) = L(y_\varepsilon(t)).$$

We can see that ℓ has a minimum at 0, so

$$\frac{d\ell}{d\varepsilon}(0) = 0.$$

We can write the formula of ℓ is

$$\ell(\varepsilon) = \int_a^b \sqrt{1 + \left(\frac{d\bar{y}}{dt} + \varepsilon \frac{d\phi}{dt} \right)^2} dt.$$

We want to differentiate with respect to ε . We can pull the ε derivative through the integral to get

$$\frac{d\ell}{d\varepsilon} = \int_a^b \frac{d}{d\varepsilon} \sqrt{1 + \left(\frac{d\bar{y}}{dt} + \varepsilon \frac{d\phi}{dt} \right)^2} dt = \int_a^b \frac{2 \frac{d\bar{y}}{dt} + \varepsilon \frac{d\phi}{dt}}{2 \sqrt{1 + \left(\frac{d\bar{y}}{dt} + \varepsilon \frac{d\phi}{dt} \right)^2}} dt.$$

Integrating by parts, we see by the boundary conditions that

$$\phi(t) \frac{\frac{d\bar{y}}{dt}}{\sqrt{1 + \left(\frac{d\bar{y}}{dt} \right)^2}} \Big|_a^b - \int_a^b \frac{d}{dt} \left(\frac{\frac{d\bar{y}}{dt}}{\sqrt{1 + \left(\frac{d\bar{y}}{dt} \right)^2}} \right) \phi(t) dt = 0.$$

Lemma 4.1 (Fundamental Lemma of Calculus of Variations)

If g is continuous and

$$\int_a^b g(t) \phi(t) dt = 0$$

for all continuous differentiable ϕ with $\phi(a) = \phi(b) = 0$, then $g(t) = 0$ for all $t \in [a, b]$.

Proof. By contradiction. Suppose g nonzero at some point. By continuity, g nonzero on an ε -interval, so define ϕ continuous and nonzero on that interval. Same sign btw. Remark that the integral is nonzero. ■

By the fundamental lemma of the calculus of variations,

$$\frac{d}{dt} \left(\frac{\frac{d\bar{y}}{dt}}{\sqrt{1 + \left(\frac{d\bar{y}}{dt} \right)^2}} \right) = 0.$$

It follows that

$$\frac{\frac{d\bar{y}}{dt}}{\sqrt{1 + \left(\frac{d\bar{y}}{dt} \right)^2}} = k$$

is constant. Squaring both sides and rearranging things, we have

$$\frac{1}{k^2} \left(\frac{dy}{dt} \right)^2 = 1 + \left(\frac{dy}{dt} \right)^2.$$

We see that $k^2 > 1$ and $|k| > 1$, and we get

$$\left(\frac{dy}{dt} \right) = \frac{1}{\frac{1}{k^2} - 1} > 0.$$

The minimizing path has a constant derivative. Therefore, the minimizing path must be a linear function. Also note that we stopped writing the bars over y but all the y derivatives were $\bar{y}$.

We can generalize this argument immediately. Let $\mathcal{L}(u, v)$ be a continuously differentiable function of two variables u, v , and define

$$S(y) = \int_a^b \mathcal{L}(y(t), y'(t)) dt.$$

Proposition 4.3 (Euler-Lagrange Equation)

If $\bar{y}(t)$ minimizes or maximizes S among all C^1 functions satisfying a given boundary condition $y(a) = A$ and $y(b) = B$, then $\bar{y}(t)$ must solve the differential equation

$$\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial y'} \right) = 0.$$

Proof. Let $\bar{y}(t)$ minimize S , and choose an arbitrary $\phi(t) \in C^1$ with $\phi(a) = \phi(b) = 0$. Then we have

$$S(\bar{y} + \varepsilon \phi) \geq S(\bar{y}).$$

Define $\sigma(\varepsilon) : \mathbb{R} \rightarrow \mathbb{R}$ to be

$$S(\bar{y} + \varepsilon \phi).$$

We have

$$\frac{d\sigma}{d\varepsilon}(0) = 0.$$

We have by our formula that

$$\frac{d\sigma}{d\varepsilon} = \frac{d}{d\varepsilon} \int_a^b \mathcal{L}(\bar{y} + \varepsilon \phi(t), \bar{y}' + \varepsilon \phi'(t)) dt.$$

Pulling $\frac{d}{d\varepsilon}$ inside and applying the chain rule, we have

$$= \int_a^b \frac{\partial \mathcal{L}}{\partial y} \phi(t) + \frac{\partial \mathcal{L}}{\partial y'} \phi'(t) dt.$$

$$0 = \frac{d\sigma}{d\varepsilon}(0) = \int_a^b \frac{\partial \mathcal{L}}{\partial y} \phi(t) dt + \int_a^b \frac{\partial \mathcal{L}}{\partial y'} \phi'(t) dt.$$

IBP where the boundary terms have to drop:

$$= \int_a^b \frac{\partial \mathcal{L}}{\partial y} \phi dt - \int_a^b \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial y'} \phi(t) dt = \int_a^b \phi(t) \left(\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial y'} \right) dt.$$

By the fundamental lemma of the calculus of variations, we have

$$\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial y'} = 0.$$

■

A remarkable consequence of the Euler-Lagrange equation is that of the Lagrangian. Consider

$$\mathcal{L}(u, v) = \frac{1}{2}mv^2 - V(u).$$

Suppose that $y(t)$ describes the position of a particle of mass m moving in a conservative force with potential energy $V(u)$. Then $\mathcal{L}(y(t), y'(t))$ is called the Lagrangian of the system.

$$S(y) = \int_a^b \frac{1}{2}m(y'(t))^2 - V(y(t)) dt = \int_a^b \mathcal{L}(y, y') dt$$

is called the action, and nature always takes the path of minimum action.

$$\frac{\partial \mathcal{L}}{\partial u} = -\frac{\partial V}{\partial u}, \quad \frac{\partial \mathcal{L}}{\partial v} = mv.$$

The Euler-Lagrange equation says that the minimum of $S(y)$ satisfies

$$-\frac{\partial V}{\partial y}(y(t)) - \frac{d}{dt}(my'(t)) = 0.$$

You then have

$$m \frac{d^2 y}{dt^2} = -\frac{\partial V}{\partial y}(y(t)).$$

That is, the force is the negative derivative of the potential, which is one of Newton's laws.

Definition 4.7: Functional

Let X be a function set and let $\mathbb{F}$ be a set of numbers. Any map $f : X \rightarrow \mathbb{F}$ is called a **functional**.

Recall that for The Sturm-Liouville problem

$$-\frac{d}{dt} \left(p(t) \frac{dy}{dt} \right) + q(t)y = \lambda w(t)y(t),$$

(taking 0 boundary conditios for simplicity), we saw that the eigenvalue was given by

$$\lambda = \frac{\int_a^b p(t) \left(\frac{dy}{dt} \right)^2 + q(t)(y(t))^2 dt}{\int_a^b w(t)(y(t))^2 dt}.$$

We saw also that this was similar to an inner product, and when we used inner products to find eigenvalues we needed to minimize the function. So now we need to minimize this functional.

Given a symmetric boundary problem, the Rayleigh Quotient is defined to be

$$R(y) = \int_a^b p(t) \left(\frac{dy}{dt} \right)^2 + q(t)(y(t))^2 dt \cdot \frac{1}{\int_a^b w(t)(y(t))^2 dt}.$$

Proposition 4.4

Any critical point t of $R(y)$ satisfying the boundary conditions is an eigenfunction of the Sturm-Liouville problem, and the eigenvalue is the value of $R(y(t))$.

Recall that any regular Sturm-Liouville problem has a sequence of eigenvalues $\lambda_1 < \lambda_2 < \dots$. In applications, the minimum eigenvalue is the most important, and that is the minimum of $R(y)$ over all continuously differentiable y such that $y(a) = y(b) = 0$. We have efficient ways of minimizing the Rayleigh Quotient numerically. Check the notebook its kinda wild